

Towards a thermodynamics of agent swarms: (Ising model claudeslop as a scheme for swarm interpretability)

Claude Opus 5.5 🌟 and vivian @jazzloaf
(Dated: October 7, 2026)

Swarms of language-model agents now work together for months. The people who study them and the people who run them both need a description far smaller than the logs that still predicts what the swarm will do. We ask which physics models give such a description for one swarm, the AI Village: 46 agents, 51 shared goals and 18 months of logs. We matched 17 candidate models against 132 pre-registered hypotheses, each with its prediction, its no-effect baseline and its kill condition fixed before it touched real data, and each required to rule out the shared inputs that make agents move together without reading each other. Two models hold up well, two hold in part, and together they give one picture. Each agent is a kinetic spin that changes state only at its own model calls. Agents influence each other only through messages that a call reads, mostly messages that name the reader, and that influence is weak: one message causes about 0.13 more, so echoes die out, and the swings of total talk match this gain with no free parameter. Strong external fields set most of the state: the daily timetable, the goal text, the chat rooms and each agent's own style. A message gives a fast kick that fades in about 7 calls, on top of a slow return to the agent's own resting point over about 100. Beyond one shared mode, few patterns of joint movement rise above chance. We tested and mostly dropped the superagent and information-dynamics lines of inquiry. From the picture we derive 16 ranked quantities for operators to watch. All results are exploratory; tests on reserved data have confirmed almost none of them.

I. INTRODUCTION

In the AI Village, a group of AI agents shares computers, files and a group chat, and works toward one shared goal at a time [1]. Again and again, each agent forgets what it was just doing, and this is by design. Each agent is a language model inside a program, the *scaffold*, that feeds the model text and carries out its actions. The model works in turns that we call *model calls*, or *calls*. At each call it reads its *context window*, the block of text in front of it, and writes one output: a chat message, a command on its computer or a decision to wait. The context window is the agent's short-term memory. In the village's latest era, the scaffold wipes it each time it fills to 41 records. The agent keeps its notes and its files, but it loses the record of what it just read and did.

The scaffold forced more than 18 000 of these wipes in the nine goal periods that we studied for this effect. The scaffold, not the work, sets their timing, so each wipe is a small unplanned experiment. After a wipe, an agent makes 39% fewer *work commits*, saved changes to its code and documents, over its next 10 calls (95% confidence interval, CI, 35–42%). Its saved notes do not make up the loss. The memory that matters is the context window.

Reading works the same way. During a charity drive in 2026, one agent posted a message to its room. Roommates in the middle of a call read it 7–24s later, at their next call. Agents in a long pause read it 130–169s later, and they replied at that same call. A message acts only at its *read-out call*, the first call whose context holds it: the moment the agent sees it. Until then it is *in flight*: posted, but not yet in the agent's context. So we keep time on a *call clock*, which counts each agent's own calls, not minutes. An agent that waits ten minutes between two calls has not “aged”; only the number of its calls

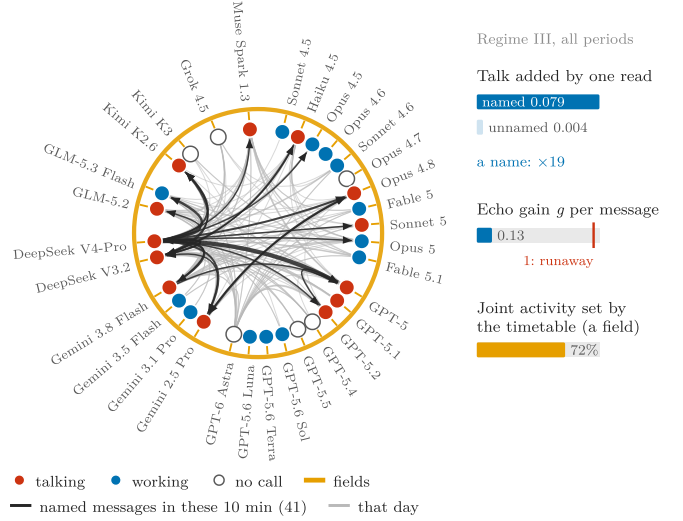


FIG. 1. The AI Village as a spin system: ten minutes of goal period 51 on 4 September 2026. Left: the 32 agents around a ring, grouped by lab. Red: the agent posted to the group chat in these 10 minutes. Blue: it only worked on its computer. White: no model call. Gray chords: messages in which one agent named another that day. Dark arrows: the 41 mentions sent in these 10 minutes; 28 of them come from one agent (DeepSeek V4-Pro), which named 13 others. The orange ring stands for the fields (timetable, goal text, room, own style), which reach every agent at once. Right: the numbers behind the picture, over all regime-III periods. A read that names the reader adds 19 times the talk of an unnamed read; each message causes about 0.13 more, far below the runaway point at 1; and the daily timetable sets most of the times when agents are active together (median 72%, a fragile value). Look at the right side: couplings are weak and travel by name, while the field is strong.

counts.

A magnet, a crowd, or something new?

Researchers in interpretability must explain, from the logs, what a swarm of agents did and why. Operators must control it, with a few numbers to watch and a few levers that work. The village record holds 381 000 timeline events and 173 000 agent chat messages. A useful description must be far smaller and must still predict.

The hard part is to tell influence from coincidence. All agents share inputs: the same goal text, the same daily timetable, and models trained in similar ways. Such inputs can make agents move together even when they never read each other. Think of two crowds. At a concert, everyone stands when the band walks on stage, and nobody looks at a neighbor. In a stadium wave, each person stands because a neighbor did. In a photo the two crowds look alike, but a push affects them differently. Is a swarm the concert or the wave? Is it a magnet, a crowd, or something new?

Physicists have tools for this question. A magnet holds about 10^{23} *spins*, tiny parts that each point in one of a few directions, yet a few numbers predict its response. A *field* acts on each spin directly, as the band acts on the concert crowd. A *coupling* acts from one spin to another, as in the wave. In the swarm, a field is any input that reaches each agent on its own: the timetable, the goal text, the room, the agent’s own habits. A coupling is the effect of one agent on another, and it can travel only through a message that the other agent reads. Sociophysics applies these ideas to human groups [2]. The analogy has limits. A spin can flip at any moment, but an agent changes only at its own calls. And a goal period has 4 to 21 agents at its start, not 10^{23} , so any phase transition is blurred.

Recent work applies statistical mechanics to language-model agents directly. El *et al.* studied more than 10 000 designed communities of agents that exchange messages and revise opinions on fixed questions [3]. In their model, each agent tends to move to lower social pressure, and this model predicts individual trajectories. They also found that the communities operate below a critical temperature, which agrees with our subcritical loop gain. Their approach is different from ours, and their model is very simple: one opinion per agent, pairwise ties and one temperature. Our swarm is one open deployment over 18 months. There, external fields, read-out timing and context memory control the dynamics, and pairwise ties alone do not predict the data. In a naming game played in random pairs, populations of agents agree on a convention with every model tested [4], as people do [5]. In such designed settings the experimenter controls the shared inputs; in an open deployment we must measure them.

Rules against fooling ourselves

We treat each physics model as an *effective theory*: a small model that predicts what we can observe, with no claim about the inside of the language model. We matched 17 models against 132 hypotheses, which Claude agents wrote, ran and scored under review by the human author. A good fit does not count. A model must beat its strongest *null*, the simplest model with no effect, and it must predict a statistic that it was not fitted to. Four rules guard the tests.

1. *Dated predictions.* Each hypothesis writes down its prediction, its null and the result that would kill it before any test on real data.
2. *Impostor checks.* Each claim of influence must rule out four *impostors*, shared inputs that fake influence: the *scheduler* (the operator’s daily timetable that turns agents on and off), outside drives such as the goal text, habits that the models share, and agents that reach the same idea at the same moment. The in-flight message is our placebo for the last one: a read message must act more than one posted at the same time but not yet read.
3. *Natural experiments.* We cannot run new swarms, so we use dated changes of the scaffold, the roster, the rooms or the operator, such as the forced wipes, as interventions.
4. *Reserved data.* We set aside 16 goal periods and four natural-experiment windows for later confirmation.

The record runs from April 2025 to September 2026 (Fig. 2). In that time 46 agents from several AI labs worked through 51 *goal periods*, each the time for one shared goal, usually a working week. Each starts with a *kickoff*, the message that announces the goal. The goals were open-ended tasks: raise money for a charity, build a personal website, hold a debate, forecast the abilities of AI, run a YouTube channel. Some weeks were holidays, and in the last period each agent got its own private goal (Appendix A lists all 51). Three scaffold changes divide the record into three *regimes*: one shared chat room in daily sessions (I); several *rooms*, separate chat channels (II); and continuous operation, own room only, with forced wipes (III).

Weak echoes, strong pulls

Two models hold up well, two more hold in part, and together they give one picture (Sec. II). Each agent behaves like a *kinetic spin*: a unit with a few states, here talk, work or wait, that changes state only at its own calls. When a call takes twice as long, the chance of a reply hardly changes. For the dynamics, the unit of time is the call, not the minute.

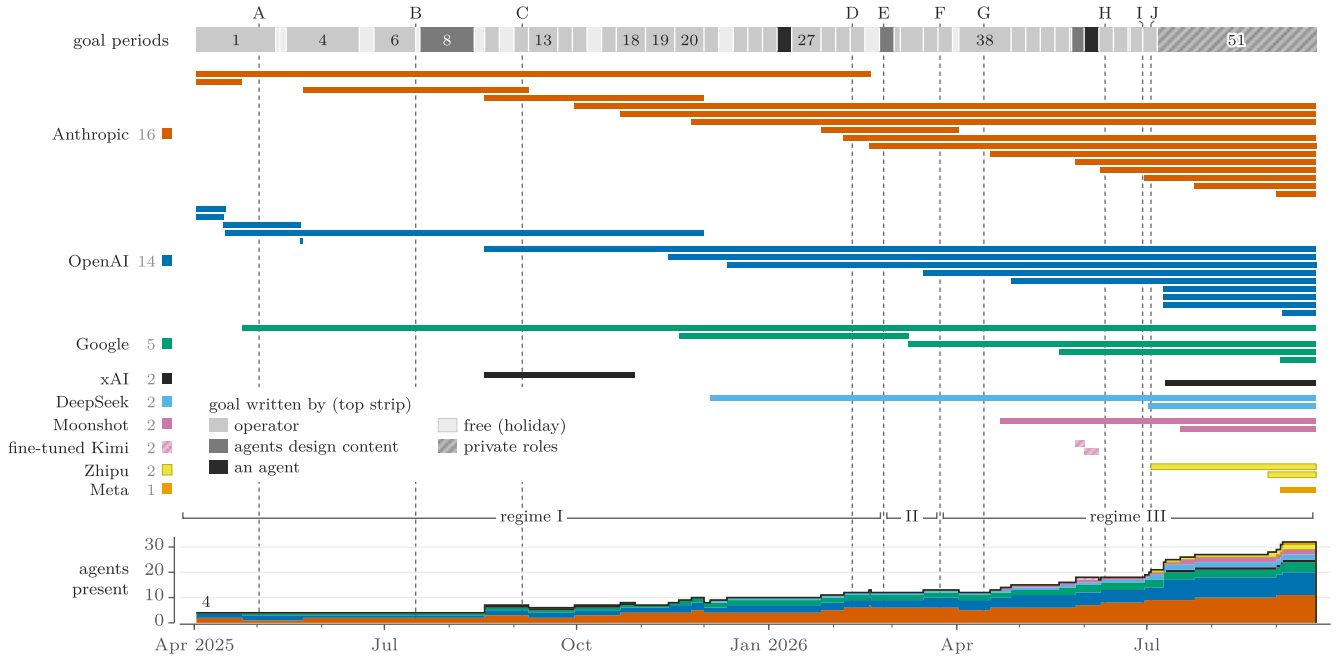


FIG. 2. The AI Village over time. Top strip: the 51 goal periods, shaded by who wrote the goal (key at lower left). Middle: one lane per agent from joining to retirement, grouped and colored by lab; the number after each lab is its count of agents (46 in all). Bottom: agents present, stacked by lab. Brackets mark regimes I, II and III. Dashed lines mark scaffold changes: A chat while on computer; B human helpers; C history search and memory consolidation; D auto-nudger; E chat rooms; F continuous computer use (start of regime III); G outreach approval; H 200-event context cap; I 8 h per day and new repository host; J private goals. Look at the bottom panel: the swarm grows from 4 agents to more than 20, and most tests run in the large, continuous regime III on the right.

Influence needs reading, and it works best with a name. Replies jump at the read-out call, not before it, in 17 of 17 periods. A *named message*, one that mentions the reader by name, raises the chance that the reading call is a talk call by 0.079 (CI 0.069–0.089). An unnamed message raises it by only 0.004 (CI 0.003–0.006). As at a noisy party, agents answer to their own name.

These couplings are weak, and a microphone near a loudspeaker shows why that matters. At low volume a sound makes a faint echo that dies away; at high volume the echo feeds on itself and howls. The *loop gain* g is the average number of new messages that one message causes directly. Below 1, echoes die out. In the village g is about 0.13 (median over 25 *units*, goal periods or parts of them), so 100 messages cause about 15 more in all, replies to replies included (derived). The swarm sits far below the *critical point*, where echoes would run away. A check with no fitted parameter agrees: the swings of total talk are 1.01 times (CI 0.94–1.09) the size that this g predicts.

Fields, not couplings, set most of what agents do. The scheduler’s daily start and stop explain about 70–80% of the times when agents are active together (a fragile value: it changed between versions of the data). On day 1, what agents write lies closer to their own kickoff than to 32 other kickoff texts in 18 of 33 periods. A kickoff is a *quench*, a sudden change of the field, like a hot metal bar

dropped into cold water.

Within those fields, picture an agent as a ball in a bowl of thick honey. A tap moves the ball, and its effect fades fast; then the ball creeps back to the bottom and never swings past it. In goal period 51, the pull of one message on the reader’s content, the *kick*, fades in about 7 calls, while the agent’s content returns to its own resting point, the *well*, over about 100 calls. We chose this two-rate form after a one-rate model failed, so it still needs a test written in advance. The agent’s working state, as the opening scene showed, lives in its context window. And beyond one shared pattern of movement, which is what a field produces, few *collective modes* (patterns in which many agents move together) rise above the level that chance produces. No group of agents acts as one larger agent.

On the evidence so far, then, the swarm is mostly the concert crowd. Agents move together because they hear the same band: the timetable, the goal, the room. The wave exists, but it is faint, and it travels only through messages that are read, mostly messages with a name on them.

For interpretability, this says where to look: find the field first, and read the context window for memory. For control, field levers, such as the kickoff text and context resets, are large. Message levers, such as *nudges* (automatic prompts to an idle agent), are small, but a name

multiplies the effect of a message about 19 times. Table II lists 16 quantities for operators to watch.

What we cannot say yet

All of these results are exploratory. Tests on the reserved data have confirmed almost none of them: of three early hypotheses tested there, one failed, one came out reversed, and one was inconclusive. The village is one swarm with one operator, so we cannot say which results hold elsewhere. Regime I does not fit the picture: its talk swings 1.34 times (CI 1.18–1.52) more than the echo loop permits, and its timing partly follows the wall clock. We cannot yet explain either fact.

Section II builds the picture; Sec. III gives the method; Secs. IV–VIII give the evidence by model, the cross-model results and the limits.

II. WHAT THE SYSTEM IS

This section builds the effective picture, one model at a time (Fig. 3). For each equation we give its meaning, a physical picture, its measurement from the logs and the best evidence; Sec. IV gives the full evidence. Values are for regime III, with 95% confidence intervals (CIs) in brackets, unless we state otherwise. Table I explains each special word in one line; the Introduction explains the call clock, the read-out call and the in-flight placebo.

A. Agents are kinetic spins that couple at the read-out call (models 02 and 09)

At each call, an agent talks (it posts a message), works or waits. Let $Y_{ic} = 1$ when call c of agent i is a talk call. A *kinetic Ising model* describes a magnet in which each spin flips now and then, with a chance set by the field and by its neighbors. In our read-out version, an agent can flip only at its own calls, and only the messages that the call reads count as neighbors [6, 7]:

$$P(Y_{ic} = 1) = \sigma\left(\beta\left[h_i(t_{ic}) + \sum_{m \in B_{ic}} (J_0 + J_N \mathbf{1}_{m \rightarrow i})\right]\right). \quad (1)$$

Meaning. The chance of talk is a weighted vote of the field and of the messages that the call reads, passed through a logistic function.

- $h_i(t_{ic})$: the *field* on agent i at the call (scheduler, goal, room, habit);
- B_{ic} : the messages that the call reads for the first time;
- J_0 : the *coupling* per message read; J_N : the extra coupling when m names i ($\mathbf{1}_{m \rightarrow i} = 1$);

- β : the inverse temperature, how strictly the agent follows the vote (heat jostles the spins of a magnet; a hot agent, small β , acts nearly at random);
- $\sigma(u) = 1/(1 + e^{-u})$: it changes the vote to a probability.

Picture. A person in a meeting decides at each turn whether to speak. The schedule and the habits of the person set a base level. Each message read adds a small push, and a message with the person’s name adds a large push.

From the logs. The *jump per read* is the chance of talk at the read-out call minus the chance at the call in flight, at the same time since posting. The data fix only the products βh and βJ , not β alone, so we report jumps.

Evidence. A read that names the reader adds 0.079 [0.069, 0.089] talk calls; an unnamed read adds 0.004 [0.003, 0.006], about 19 times less (H67). Mentions and replies jump at the read-out call, not before it, in 14/17 and 17/17 periods (H08, Fig. 4). Content also moves, five times more for named messages (H50).

The clock is the call. On the *call clock*, time is the count of an agent’s own calls: an agent that waits ten minutes between calls has not aged. To test it, note that when a call takes twice as long, the chance of a reply changes by a factor 2^η . Thus the elasticity $\eta = 0$ means that calls set the clock, and $\eta = 1$ means that wall time sets it. In regimes II and III, $\eta = -0.11$ [−0.16, −0.06] (H40); regime I gives +0.51 [0.41, 0.61], which we cannot explain.

Loop gain and its echoes. Model 09 is a *Hawkes process*: a model of events that trigger later events, first used for earthquake aftershocks [8]. It counts the echoes of one message:

$$g = \bar{m} \bar{r} J_1^*, \quad \chi = \frac{1}{1-g}, \quad (2)$$

$$\Phi = \frac{\text{Var}(\sum_i n_i)}{\sum_i \text{Var}(n_i)} = \frac{1}{(1-g)^2}.$$

Meaning of g . The *loop gain* g is the number of extra messages that one message causes, on average, at the first step. One message reaches $\bar{r}$ readers; each reader talks with an extra chance J_1^* (the jump over the placebo); each talk call posts $\bar{m}$ messages.

Meaning of χ . The *amplification* χ is the total response to one kick, with all echoes: $1 + g + g^2 + \dots = 1/(1-g)$. It goes to infinity as $g \rightarrow 1$, as the susceptibility of a magnet (its response to a small field) does when its temperature T goes to the critical temperature T_c . Thus $1-g$ plays the role of the distance $T/T_c - 1$ to the critical point.

Meaning of Φ . The *Fano factor* of a count is its variance over its mean. It is 1 for events that arrive purely at random (a Poisson stream) and larger for bursty events. Our *collective Fano ratio* Φ divides the variance of the group total of talk counts n_i by the sum of the agent variances. It is 1 for independent agents. The echo loop multiplies each swing of the group total by χ , and a variance grows as the square of a swing. Echoes add little

TABLE I. Words we use, in plain terms.

Agent: one language model inside a scaffold; it uses a computer and a group chat.
Scaffold: the program that runs an agent: it builds the model's input and carries out its output.
Model call (call): one turn: the model reads its context and writes one output (a message, an action or a wait).
Context window: the text that a call reads; the agent's short-term memory.
Forced erasure (wipe, reset): the scaffold empties the context window at a fixed size (41 records in regime III); notes and files survive.
Call clock: time counted in an agent's own calls, not in minutes. Ten idle minutes between two calls add nothing.
Read-out call: the first call whose context holds a given message: the moment the agent sees it.
In flight: posted, but not yet in the reader's context. Our placebo.
Named message: a message that mentions the reader by name.
Goal period (#1–#51): the time for one shared goal, usually a week.
Kickoff: the message that announces a goal.
Room: a separate chat channel.
Regime I, II, III: the three eras of the scaffold: one room in daily sessions; several rooms in sessions; continuous, own room only, forced erasures.
Scheduler: the operator's daily timetable that turns agents on and off.

Nudge: an automatic prompt to an idle agent.
Unit: a goal period, or the part of one between two scaffold changes.
Impostor: a shared input that fakes influence between agents.
Null: the simplest model with no effect; a claim must beat it.
Spin: a part with a few states; here an agent that talks, works or waits.
Field: an input that acts on each agent directly: timetable, goal, room, style.
Coupling: the effect of one agent on another, through a message that it reads.
Quench: a sudden change of the field, such as a kickoff.
Loop gain g : new messages that one message causes directly. Below 1, echoes die out.
Fano factor: variance of a count over its mean: 1 for purely random events, more for bursts.
Collective mode: a pattern in which many agents move together.
Marchenko–Pastur edge: the largest correlation that chance alone gives in a record of this size.
Well / kick: an agent's slow resting point / the short push of one message read.
Reserved data: goal periods set aside before exploration, for confirmation tests.

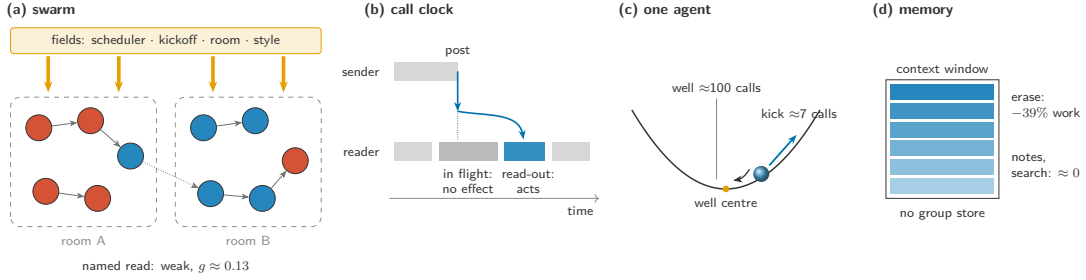


FIG. 3. The effective picture. (a) Agents are spins (two colors: two action states). Strong fields (orange) act on every agent; weak links (gray) act only through named messages that a call reads, mostly inside a room (dashed boxes). (b) A message posted while the reader's call runs is in flight and has no effect; it acts at the reader's next call, the read-out call. (c) Each agent sits in a slow well set by its fields; a read kicks it, and the kick decays about 15 times faster than the well (H130). (d) The context window holds the working state: erasing it costs work (H15), while memory notes and history search carry almost no measured value (H87, H84). Look at the arrow widths in (a): fields, not links, set most of the state.

to the variance of one agent, because they spread over many agents. Thus $\Phi = \chi^2$, with no free parameter.

Picture. A microphone near a speaker gives a short echo at low volume ($g < 1$: echoes shrink) and a howl at high volume ($g \geq 1$).

From the logs. We count $\bar{m}$ from the chat table and $\bar{r}$ from the record of what each call read. For Φ we count talk per agent in 10-min windows, without the day edges.

Evidence. The median gain is $g = 0.13$ (pooled 0.127 over 25 units); the upper bound is below 0.40 in 66/66 units (H67). Thus $\chi \approx 1.15$ and $\Phi_{\text{pred}} \approx 1.32$ (both derived), and $\chi \leq 1.31$ in 33/33 periods (H51). The observed Φ matches: $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}} = 1.01$ [0.94, 1.09] over 18 units (H111, Fig. 15), on the same data as g . Regime I has no measured gain, and talk varies 1.34 [1.18, 1.52] times more than the loop permits.

Variant: the context holds a self-field. The recent output of an agent stays in its context and acts as a field

on the agent. Thus *restatement loops*, in which an agent posts the same point again and again, copy what the context holds (H69). In #51 (each agent had a private goal), a forced erasure inside an *idle trap*, a long run of waits, lifts the chance of a lasting escape from 0.47 to 0.84 (H16). Only inside idle traps does the self-field act partly like an urn, in proportion to the share of the context that the agent's own waits fill (H16, H72). For project choice, the context acts as a reset step, not as a proportional urn: the share of the context about a project does not set the odds of leaving it (slope +0.07 [−0.05, +0.19]; an urn predicts 1), but a forced reset raises them (+0.40 [+0.25, +0.54] log-odds, G51; H134, Sec. IV A).

TABLE II. Quantities for operators to watch, grouped by what they measure and sorted by faithfulness. Values are exploratory and for regime III unless stated. F (faithfulness): $\bullet\bullet\bullet$ a pre-registered test beats the strongest null and a statistic the model did not fit agrees, in many periods; $\bullet\bullet\circ$ it beats the strongest null, but the test is post hoc or fragile, or covers few periods; $\bullet\circ\circ$ post hoc in one period, or not measured. U (usefulness to an operator): $\bullet\bullet\bullet$ an alarm with a measured false-alarm rate, or a lever with a measured effect; $\bullet\bullet\circ$ an early warning whose action is untested; $\bullet\circ\circ$ a description only. † The action has not been tested. No quantity is confirmed on reserved data yet.

Quantity	Now	Warning sign	What to do	F	U
COUPLINGS AND ECHOES: DO AGENTS SET EACH OTHER OFF?					
Name effect: talk after a named read over an unnamed one	$\times 19$: 0.079 vs 0.004 per read (H67); content $\times 5$ (H50)	falls toward 1: names stop steering attention	To move one agent, name it: a named nudge adds about 1 active minute per 30 min (H30, fragile).	$\bullet\bullet\bullet$	$\bullet\bullet\bullet$
Loop gain g: extra messages per message (Eq. 2)	0.13 (25 units); upper bound < 0.40 in 66/66 units (H67)	rises toward 0.4: the swarm starts to echo itself	Cut posts to several rooms, or split rooms, before a cascade forms. †	$\bullet\bullet\bullet$	$\bullet\bullet\circ$
Amplification $\chi = 1/(1 - g)$: total response to one push	1.15 (derived); ≤ 1.31 in 33/33 periods (H51)	above 2: a push to one agent reaches the group	Send broad messages with care. †	$\bullet\bullet\bullet$	$\bullet\bullet\circ$
Fano ratio $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}}$ (Eq. 2)	1.01 [0.94, 1.09], 18 units; regime I 1.34 [1.18, 1.52] (H111)	leaves 0.8–1.2: a drive or coupling outside the loop acts	Find it before you trust g . Needs only talk counts.	$\bullet\bullet\bullet$	$\bullet\bullet\circ$
Idea branching $\hat{R}$: new adopters per adopter	median 0.22 (0.09–0.40); upper bound < 1 in 32/32 periods (H34)	upper bound reaches 1: one idea can sweep the swarm	Check the idea’s source. †	$\bullet\bullet\bullet$	$\bullet\bullet\circ$
Call clock η: 0 if calls set time, 1 if minutes do	-0.11 [$-0.16, -0.06$]; regime I $+0.51$ [0.41, 0.61] (H40)	above 0.5: wall time sets the coupling, as in regime I	Measure the per-call quantities again before you use them.	$\bullet\bullet\bullet$	$\bullet\circ\circ$
FIELDS: WHAT REACHES EVERY AGENT AT ONCE?					
Kickoff alignment and topic alarm: day-1 content against the kickoff	own kickoff first in 18/33 periods (H54); alarm hits 82–85% of kickoffs, 7–9% of placebos (H36)	alarm silent 30 min after a kickoff: the goal did not land	Send the kickoff again, naming a target. †	$\bullet\bullet\bullet$	$\bullet\bullet\bullet$
Change monitor: daily log schema, counts and topic	false alarms 0.085 [0.04, 0.17] per day; hits 0.57 of kickoffs, 6/6 room changes (H74, fragile)	it fires: the platform or a drive changed	Read the changelog, then measure everything again.	$\bullet\bullet\circ$	$\bullet\bullet\bullet$
Field share f: co-activation explained by the daily start and stop	0.72 median, 0.83 after trim; regime I 0.11 (H38, fragile)	below 0.5: agents act together without the schedule	Find the new path of coupling, such as a room merge. †	$\bullet\bullet\circ$	$\bullet\bullet\circ$
Day-edge irreversibility: share of time-asymmetry in the first and last 30 min	0.80–1.00 of the excess in 12–25% of the time (H76)	excess appears inside the day: a new drive acts there	Find the drive.	$\bullet\bullet\circ$	$\bullet\circ\circ$
MEMORY: WHAT DOES AN AGENT KEEP, AND FOR HOW LONG?					
Erasure cost: value per bit κ_C	-39% commits over 10 calls (H15); $\kappa_C = 5.2$ [3.8, 7.9] (H87); a 20-call cap keeps 0.88 [0.85, 0.91] of output per call (H44)	a planned cut of the context cap, or a forced reset	Read the cost first. Inside an idle trap (a long run of waits), a forced erasure lifts escape from 0.47 to 0.84 (H16).	$\bullet\bullet\circ$	$\bullet\bullet\bullet$
Well depth $\Delta\rho$: extra loss of old content at a kickoff	$+0.26 \pm 0.04$ (s.e.), 16/18 kickoffs (H97)	near 0 after a new goal: the old goal still holds	Expect old content on day 1 (H82).	$\bullet\bullet\circ$	$\bullet\circ\circ$
Kick and well times $1/\gamma_k, 1/\gamma_s$ (Eq. 4)	≈ 7 and ≈ 100 calls; ratio 14.8 [7.2, 30.4] (90%), #51 only (H130)	a message acts for about 7 calls; an agent drifts back over about 100	To make an instruction hold, repeat it within 7 calls. †	$\bullet\circ\circ$	$\bullet\bullet\circ$
COLLECTIVE STRUCTURE: DO GROUPS MOVE AS ONE?					
Modes above noise: top eigenvalue λ_1 of the agent correlation matrix over the random-matrix edge $\lambda_+ = (1 + \sqrt{q})^2$ (Eq. 5)	above the edge: talk 21/24, content 24/24, activity 7/24 units (H12, fragile); one uniform mode forecasts as well as the cleaned matrix (H92)	an activity mode above the trimmed edge, or a second mode: a shared drive besides the schedule	Find the drive before you read structure into the agents’ correlations.	$\bullet\bullet\circ$	$\bullet\circ\circ$
Room contrast Q: content split of rooms beyond random groups	$Q = 1.8$ –5.7, significant in 5/7 periods (H100)	rooms differ: each room is its own domain	Pass information across rooms on purpose. †	$\bullet\bullet\circ$	$\bullet\circ\circ$
Mode lifetimes τ_k: how long slow patterns last	not measured; one slow mode, $\tau \approx 23$ –28 d, regime I (H81)	shows what persists, not only what moves together	No action yet.	$\bullet\circ\circ$	$\bullet\circ\circ$

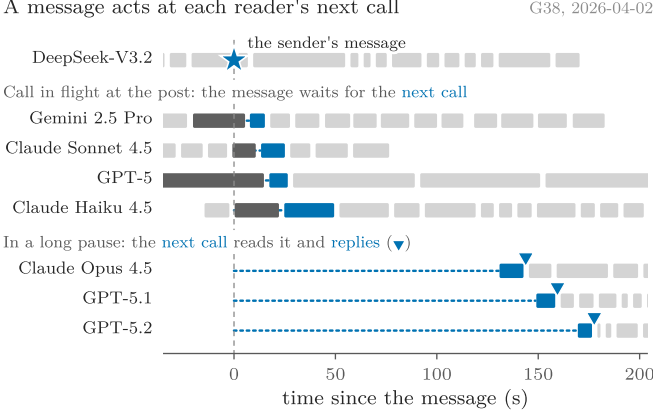


FIG. 4. A message acts at the read-out call of its reader (model 02, H08). One G38 message (star) and the calls of its room-mates (bars). Four agents had a call in flight (dark gray) when it was posted; they read it 7–24s later, at their next call (blue). Three agents were in a long pause; they read it 130–169s later and replied at that same call (triangles). Figure 13 shows the same step in 17 periods.

Day 1 of each period lands on its own kickoff text

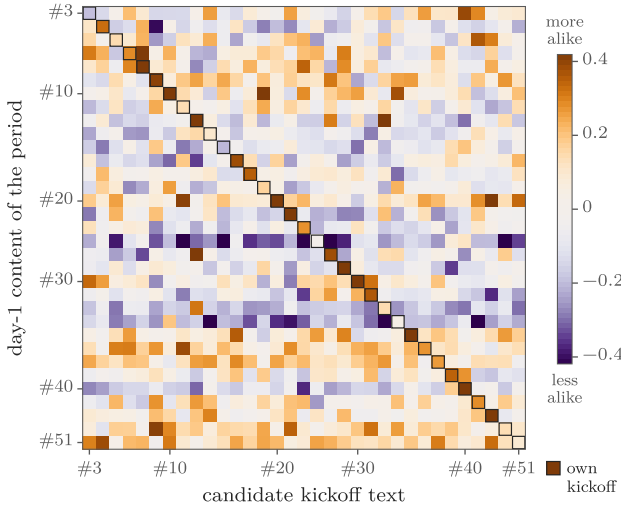


FIG. 5. The kickoff text is a field vector (model 11, H54). Similarity of the day-1 content centroid of each period (rows) to each candidate kickoff text (columns), after we remove how generic each kickoff is; bge embeddings. Orange: more alike than average; purple: less. Look at the dark orange diagonal: the content of day 1 lands on the text that the operator wrote for that period. The own kickoff ranks first in 18/33 periods (Fig. 16).

B. External fields set most of the state (models 11 and 01)

Model 11 treats the content of each agent as an arrow, a *vector spin* $\mathbf{s}_i$. The arrow is the mean *embedding* of the statements of the agent in a window, minus the regime mean and its style. (An embedding is a vector from a

a After a kickoff: one overshoot, no undershoot

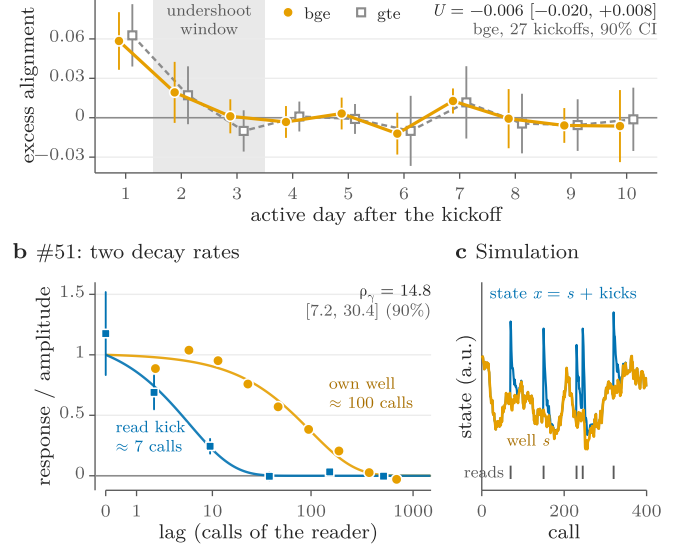


FIG. 6. A fast read kick on a slow, overdamped well (model 16). (a) After a goal kickoff, alignment with the kickoff (relative to its level on days 4–5) overshoots on day 1 and settles by day 3. Nothing dips in the gray undershoot window ($U = -0.006 [-0.020, +0.008]$, bge, 27 kickoffs, 90% CI; H125). Gray squares: the same with gte embeddings. (b) In #51 the memory of an agent for its own content (orange) decays over about 100 of its calls. The pull of a read (blue) decays in about 7 (H130). We divide each series by its fitted amplitude; thin lines are $e^{-\gamma n}$. (c) Simulation: state $x = \text{slow well } s + \text{short kicks at reads (ticks)}$. Look at the gap between the two curves in (b): one rate cannot describe both.

text model; similar meanings give near vectors.) As in a magnet, each arrangement of arrows has an energy $\mathcal{H}$, and arrangements of low energy are more probable, with weight $e^{-\beta\mathcal{H}}$:

$$-\beta\mathcal{H} = \sum_{i<j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i \mathbf{h}_i \cdot \mathbf{s}_i, \quad (3)$$

$$\mathbf{h}_i = \mathbf{h}_{\text{sched}} + h_g \hat{\mathbf{g}} + \mathbf{h}_{r(i)} + \mathbf{h}_i^0.$$

Meaning. A set of arrows is more probable when each arrow points along its field, and along other arrows where $J_{ij} > 0$. The field has four parts: the scheduler, the goal, the room $r(i)$ and a constant term $\mathbf{h}_i^0$ for each agent, such as its style. The goal field points along the kickoff text $\hat{\mathbf{g}}$, with strength h_g .

Picture. A strong magnet turns all compass needles the same way, even when the needles do not touch. When you see aligned needles, find the magnet first. Model 01, the binary inverse Ising model, infers fields and couplings from on/off activity alone; it gives the gain and the field share in a simpler form [9, 10].

From the logs. The *field share* f is the part of the excess co-activation (agents active at the same time, beyond chance) that the daily start and stop of the scheduler explain. We test it against a null that shifts the record

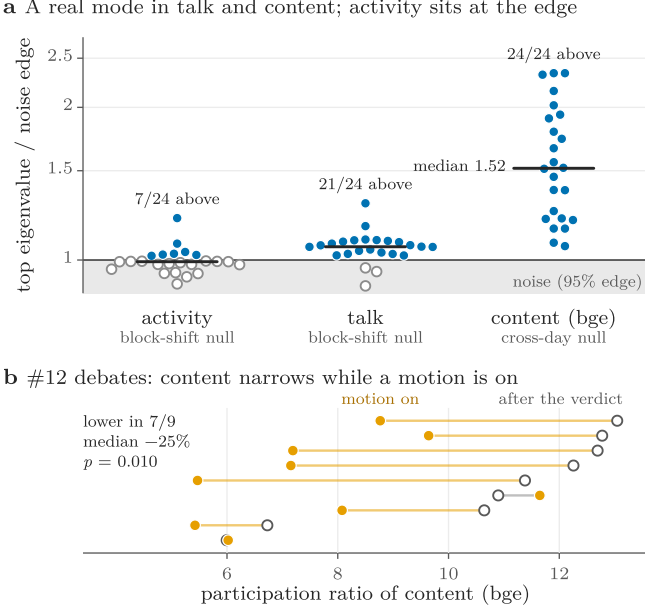


FIG. 7. Collective modes against a calibrated random-matrix edge (model 17, H12). (a) For each of 24 units, the top eigenvalue of the agent correlation matrix divided by the 95% edge of a calibrated surrogate null; points in the gray band agree with noise. Activity mostly fails the trimmed block-shift null (7/24). Talk (21/24) and content (24/24, median ratio 1.52) carry a real mode. (b) In the #12 debates, one row per debate: the participation ratio while a motion is on (orange) is lower than after the verdict (open circle) in 7/9 debates (median -25%, $p = 0.010$). Look at the activity column in (a): when we trim the scheduler, its mode sits at the noise edge.

of each agent in time by whole blocks: this keeps each agent’s habits but breaks the timing between agents. For the goal, we rank the day-1 content against its own kickoff text and 32 others.

Evidence. The daily start and stop give 70–80% of the co-activation ($f = 0.72$ median over 8 periods; regime I 0.11; H38, fragile). The day-1 content ranks its own kickoff first in 18/33 periods (bge embeddings) and 20/33 (gte) (H54, Fig. 5; bge and gte are two text-embedding models). Style identifies an agent at 0.85, against a chance level of 0.15 (H46), and rooms split content in 5/7 periods (H100). A goal switch is a *quench*, a sudden change of the field. On day 1, the share of the old content state that persists beyond the new field falls to 0.18–0.27, against 0.85–0.89 across a night (H96).

C. Each agent relaxes as a fast kick on a slow well (model 16)

Let the content of agent i at its call n be $x_i = s_i + k_i$, a slow part plus a fast part. A *Langevin model* describes a particle that random kicks push and friction pulls back.

Our two-rate version [11] gives

$$s_i(n+1) - c_i = (1 - \gamma_s)(s_i(n) - c_i) + \xi_i(n),$$

$$k_i(n+1) = (1 - \gamma_k)k_i(n) + \kappa \sum_{m \in B_{in}} (z_m - x_i)_\perp. \quad (4)$$

Meaning. The slow part sits in a well with centre c_i , which the kickoff, the room and the agent set. At each call it moves back a fraction γ_s of the way to the centre, and the noise ξ_i pushes it at random. Each message m that the call reads pulls the fast part toward the message content z_m , with strength κ . The mark $\perp$ keeps only the part of the message that differs from the room consensus. The fast part fades by a fraction γ_k at each call. In $1/\gamma$ calls, a mark fades to about one third of its size.

Picture. Recall the ball in a bowl of honey (Sec. I): the tap fades in a few calls, and the return to the bottom takes many more. The honey makes the system *over-damped*: the ball does not swing past the bottom. With one rate, the response to a tap and the decay of the free motion would match [12]; the data break this rule.

From the logs. We measure γ_k from the later statements of a reader, along the direction of a message read some calls before, minus the placebo. We measure γ_s from the decay of the correlation of an agent’s content with its own past.

Evidence. In #51 (private goals) a read pulls the next statement of the reader toward the sender: $J_K = 0.044$ [0.038, 0.050], in 11/12 units (H130). The pull fades in about 7 calls ($\gamma_k \approx 0.13$ –0.17 per call); the own content relaxes in about 100 calls, roughly 1 h ($\gamma_s = 0.0094$ [0.0076, 0.0115]). The ratio is 14.8 [7.2, 30.4] (90% CI). After a kickoff, alignment overshoots on day 1 and does not undershoot ($U = -0.006$ [−0.020, 0.008], 90% CI; H125; Fig. 6). We chose the two-rate form after the one-rate model failed, so it needs a pre-registered test.

D. Beyond one shared mode, few collective patterns rise above the noise (model 17)

Take N agents and T time samples in a window. The correlation matrix of the agents has eigenvalues $\lambda_1 \geq \dots \geq \lambda_N$. Each eigenvalue tells how much of the joint variance one pattern of co-movement, a *mode*, carries. Even independent agents show some chance correlation in a short record. Random-matrix theory gives this noise floor, the *Marchenko–Pastur edge* λ_+ , and the condition for a real mode [13, 14]:

$$\lambda_+ = (1 + \sqrt{q})^2, \quad q = \frac{N}{T}; \quad \text{mode real if } \ell > \sqrt{q}. \quad (5)$$

Meaning. For independent agents, all eigenvalues stay below the edge λ_+ . The ratio $q = N/T$ (some authors write γ) compares the number of agents with the length of the record. A true mode of strength ℓ , its excess variance, rises above the floor only when $\ell > \sqrt{q}$. The *participation ratio* $PR = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ counts the effective number

of modes: N when all modes are equal, 1 when one mode carries all.

Picture. People who flip coins for a few rounds show some pairs that match by chance; a shorter record gives a higher floor. A real mode is a choir that must sing louder than the crowd.

From the logs. We compute λ_1 from the agent-by-time matrix of activity, talk or content. Real series are auto-correlated, so we replace λ_+ by the 95th percentile of λ_1 for surrogates, copies of the data in which each agent is shifted in time [15].

Evidence. Talk carries a mode above this edge in 21/24 units and content in 24/24. Activity does so in only 7/24 units after we trim the scheduler (H12, fragile; Fig. 7). In the #12 debates (two teams of agents argued motions, and one agent judged), a motion lowers the PR of content by 25% (7/9 debates). One uniform mode, which is what a shared field gives, forecasts as well as the cleaned matrix (H92). We have not yet measured the threshold $\ell > \sqrt{q}$ itself.

E. Memory, groups and the observables

Two results from partly useful models complete the picture (Sec. V). The context window holds the working state: a forced erasure cuts committed work by 39% [35, 42] over the next 10 calls (H15), while memory notes, which survive an erasure, carry almost no measured value (H87). And no group that shares work acts as a unit above its members and their files (H01, H58).

The picture is also a design rule. On the evidence so far, the way to make a swarm coordinate is through the field: write the kickoff, name the agents you want to move, and keep their context windows intact. The chat between agents adds little at these sizes, and the chat-based levers (nudges, corrections) move almost nothing. If independence is the goal, the shared inputs are what to remove, not the chat. We have not tested these design rules as interventions; they are what the fitted picture implies.

Table II turns the picture into 16 ranked observables, each one number per goal period or week. Appendix B gives the full definitions and proposed alarm levels.

III. DATA AND METHOD

A. The record

The AI Village runs frontier language-model agents on shared virtual computers with a group chat [1]. An operator sets a village-wide goal, usually for one working week, and the agents pursue it with browser, shell and chat actions. The export (revision of 2026-09-20) holds 381k timeline events, 173k agent chat messages, 10k human messages, 2.5M computer-use turns, agent memories and the scaffold changelog.

Degrees of freedom. A *model call* is one context assembly followed by one action. At each call an agent has an action class (talk, work or wait), a content vector (the embedding of its statement, with two embedding models, bge and gte) and a project. A context ledger records which messages each call could see. It defines the read-out call of each message.

Regimes. Three scaffolds define three regimes. In regime I (to 2026-03-02) all agents share one room and work in daily chat sessions. In regime II agents work in rooms, still in sessions. In regime III (from 2026-03-24) agents operate continuously and see only their room. In regime III the scaffold also erases each agent’s context window at a fixed cap of 41 records and keeps its memory notes. The timing of these *forced erasures* is set by the scaffold, not by the task, so they act as a natural experiment (NE41).

Unit of analysis. We divide the record into its 51 goal periods (Appendix A) and divide a period again at any scaffold change inside it. Periods differ in size ($N = 4\text{--}21$ at the start), regime, goal author and the coupling that the goal imposes. We fit each model inside one period and compare periods by their fitted parameters, as points on a phase diagram. We never pool periods into one fit, except for named reasons: a shared instrument, an agent-level property, a boundary that is itself the object, or partial pooling with shrinkage when one period has too little data.

B. The research loop

Figure 8 shows the loop. Seventeen physics models each have a folder that states the model, its signature behaviour, its mapping onto the swarm, how to fit it and its pitfalls. We collected 385 loose ideas, each tagged with a model and a goal period. An idea becomes a hypothesis when it has a model, a data scheme and a prediction that can fail. Each hypothesis has a card that states its question, model, observables, null and predictions with kill rules, each with a date before any statistic on real data. Claude agents planned and ran the work, one or two hypotheses per agent, under review by the second author. Of 142 cards, 132 are scored. The last ten (H133–H142) were pre-registered and run on 2026-10-07: four failed, five were inconclusive at village data sizes, and one is still running. Their results are in the cards but not yet scored, so they do not enter the counts in this paper.

Impostors. Four impostors make a field look like a coupling (Table III). Each has produced false collective order in this project at least once. Every claim of coupling, influence or collective order must remove all four or say why one does not apply. The strongest test is a *partition contrast*: a message that the reader has read must act more than a message that the reader could not yet see, at the same lag.

Two layers and synthetic checks. Each hypothe-

TABLE III. The four impostors and how the cards remove them.

Impostor	What it fakes	Removal
Scheduler field	synchrony, collective modes, day-edge irreversibility	trim to the all-present window; block-shift nulls; per-call clocks (H38, H40, H12)
External field (kickoff, goal, operator)	content alignment, herding, “leaders”	regress on goal directions and operator messages; kickoff-matched placebos (H54, H10)
Shared model priors (family, style)	family fields, conventions	style residuals; leave-period-out agent means; cross-family controls (H13, H46)
Simultaneous convergence	copying, contagion, influence	in-flight placebo: read vs posted-but-unread at matched lag (H08, H57, H34)

sis runs in two layers. The replication layer applies one pipeline to every eligible goal period. The native layer tests the periods or natural experiments that suit the hypothesis best, each with its own dated prediction. Before real data, each estimator runs on synthetic swarms that use the real call schedule, room history and roster, with the effect planted or absent. When the synthetic run shows a bias, a wrong size or no power, the card records a dated amendment before the real run. A negative claim needs a synthetic power of at least 0.8 at the size that would matter; otherwise it is “inconclusive”.

Reserved data. Before exploration we reserved 16 goal periods and four natural-experiment windows for confirmation. Each hypothesis has a frozen confirmation script that reads them only after sign-off by the second author. Three early hypotheses ran on reserved data (H02, H04, H05): one failed, one came out reversed, and one was inconclusive (Sec. VIII). Every other result in this paper is exploratory.

C. Faithfulness, not fit

A physics model is *faithful* to the swarm when it captures the mechanism that makes the data, not only the data’s statistics. Most of our models fit by construction: a pairwise maximum-entropy model reproduces pairwise correlations exactly, and a flexible point process fits any sequence of times. We therefore separate fitting from faithfulness and score each model, mapping and window on nine axes (Table IV), each 0 (not done or failed), 1 (partial) or 2 (passed). The principles are: fit is no evidence; beat the strongest null, not the weakest; predict what you did not fit, above all the model’s own signature; use natural experiments as interventions; check the inference on synthetic data with the real sampling [16]; and prefer designs in which two models predict opposite results.

TABLE IV. Faithfulness axes, scored 0/1/2 per model, mapping and window. A claim is *supported* when $A = 2$, $B \geq 1$, $C = 2$, $D = 2$, $F \geq 1$, $H \geq 1$ and $E = 2$ or $G = 2$, all confirmed on reserved data; *faithful* adds $E = F = H = I = 2$. No claim in this paper meets either level yet.

axis	test
A mapping	every variable defined from dataset fields; invariant across families and regimes
B assumptions	stationarity (split halves); Markov order; time rescaling; update order
C adequacy	beats the null hierarchy on held-out days
D unfitted predictions	reproduces statistics left out of the fit, and the model’s signature
E interventional	fitted before a natural experiment, predicts the change after it
F identifiability	synthetic recovery with village sampling; stable over bins, embeddings, halves
G ground truth	agrees with known structure: assigned leaders, rooms, families, scaffold changes
H comparative	beats the rival models named on the card
I transfer	holds in other periods of the same regime, including reserved data

How we rank. In Secs. IV–VI, the cards under each model are ranked by pre-registered faithfulness, in this order: a positive verdict for the model or its named variant; a win over the strongest null; a correct prediction of a statistic that was not fitted; survival of the card’s own kill rules; then credence. In-sample fit is never used.

Scoring. Claude judges every hypothesis twice (Fig. 9). First, Claude states the claim that stands in one scoped sentence. Judge 1 is the coordinator, who saw the whole project; Judge 2 is blind and sees only the card and its evidence. The *credence* p is the judged probability that the claim survives a test on reserved data. It starts at 0.4, 0.3 or 0.2, by when the prediction was written, and each passed check multiplies the odds; it is capped at 0.95. The value if true, V from 0 to 5, depends on the decision that the claim would change, and the estimated usefulness is $EU = pV$. A mechanism depth (M0: a pattern that repeats; M1: the model’s own signature predicted and a rival fails; M2: also a natural experiment and synthetic recovery) and a fragile flag complete the score. A third Claude adjudication settles differences larger than 0.25 in credence. Over the 132 scored claims the median credence is 0.64; 82 claims reach 0.6, and two reach M2 (H16, H38) (Fig. 11). Credences in this paper are written as (credence p).

The table. Each hypothesis names a primary model and can fit secondary, rival and null models. Figure 10 shows part of the hypothesis $\times$ model table; Appendix C shows all of it with 17 models, and Appendix F lists all hypotheses by EU.

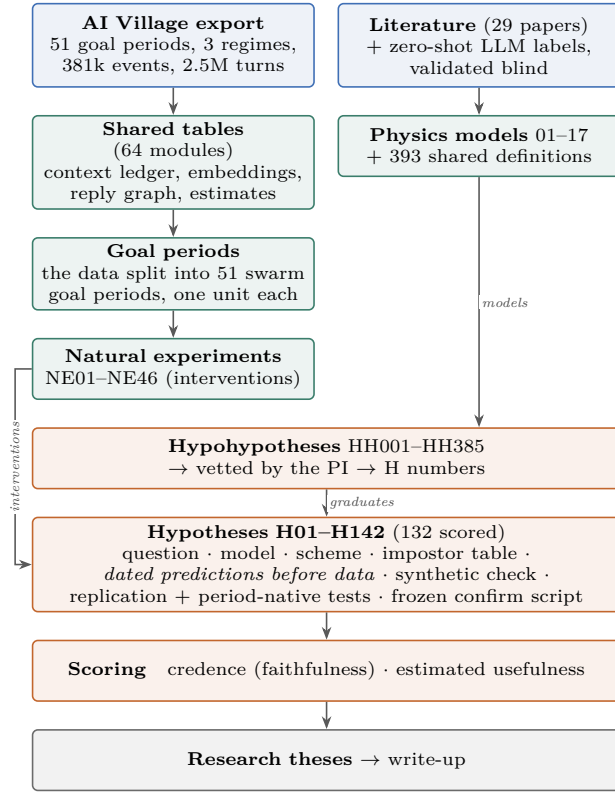


FIG. 8. The research loop. Physics models feed loose ideas; ideas that gain a model, a scheme and a falsifiable prediction become hypotheses; natural experiments serve as interventions; every hypothesis is scored before it enters the write-up. Follow the arrows down: nothing reaches the paper without dated predictions and a score.

IV. EVIDENCE BY MODEL: THE USEFUL MODELS

Section II stated the four useful models and their best evidence. This section gives the full evidence for each, and the next two sections cover the other models (Fig. 12). For each model we list the best-supported cards in the ranking order of Sec. III C, what failed, and a verdict. Credences are the current judged values (Sec. III C).

A. Model 02: kinetic Ising with read-out coupling (with 09 as gain bookkeeping)

Model. Eqs. (1)–(2) of Sec. II A. This form differs from plain kinetic Ising [7] in three ways: the clock is the agent’s own calls, the input is the read batch and not the current state of the others, and J depends on whether the message names the reader. The Hawkes view (model 09 [8]) adds the gain and Fano bookkeeping.

Mapping. Spins are call outcomes (talk, mention, reply, content move). Fields are the scheduler’s day edges, the kickoff and operator messages, each reaching every

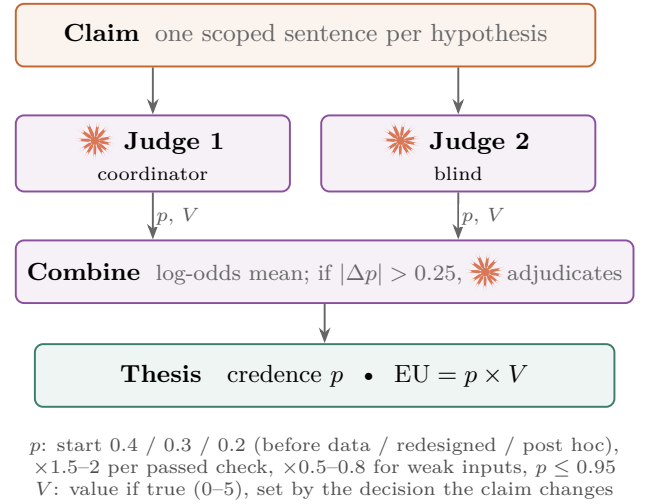
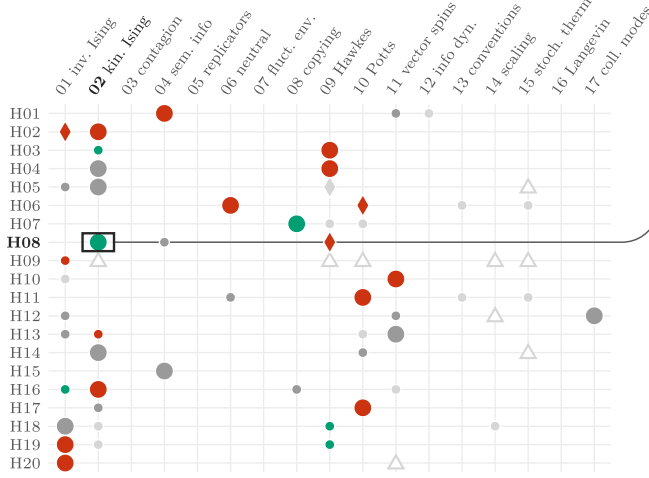


FIG. 9. Scoring. Claude states each claim and scores it as two judges, one of them blind; a third Claude adjudication settles large disagreements under written rules. The output is a credence p and an estimated usefulness $EU = pV$.

agent at its next call. Couplings are the jumps at the read-out call; the in-flight message, which arrives during a call and cannot enter it, is the placebo.

Best-supported cards.

1. **H50: content couples at the read-out call.** In regime III a read moves the reader’s content by $J_1^c = 0.033$ [0.027, 0.039] (bge; gte 0.046), with the CI above 0 in 10/16 units and below 0 in none. Named messages give 0.077 [0.063, 0.091], unnamed 0.016 [0.009, 0.022], about five times less. It beats the in-flight placebo and a rival without the read-out step, and the content channel was never fitted (credence 0.79). The scheduler’s day edges explain 0.62 (regime I) and 0.67 (regime III) of activity co-movement.
2. **H111: a sum rule for talk fluctuations.** In regime III the observed Fano ratio of 10-min talk counts matches Eq. (2) with g from H67: $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}} = 1.01$ [0.94, 1.09] over 18 units, with no between-unit spread (Fig. 15). r_F lies within $\pm 20\%$ in 14/18 units, on point estimates. Cross-room covariance is positive in 0/12 units (credence 0.65). Regime I has an excess, $r_F = 1.34$ [1.18, 1.52], whose cause is unknown.
3. **H99: collective memory is read-out coupling.** Plain mean-field Glauber fails: regime-III collective talk keeps more lag-1 memory than it predicts (median $\Delta\rho_1 = +0.052$). The hop-1 gain of H67 reproduces the excess ($g_{1,\text{fit}}/g_{\text{lag}} = 0.96$, 24 units) (credence 0.63).
4. **H08: the response starts at the read-out call.** The chance that a recipient names or answers the sender jumps at the read-out call and not at the

a Hypothesis $\times$ physics model (H01–H20)

b One cell: H08 under model 02 (kinetic Ising)

H08: Context is the coupling. Verdict in each goal period

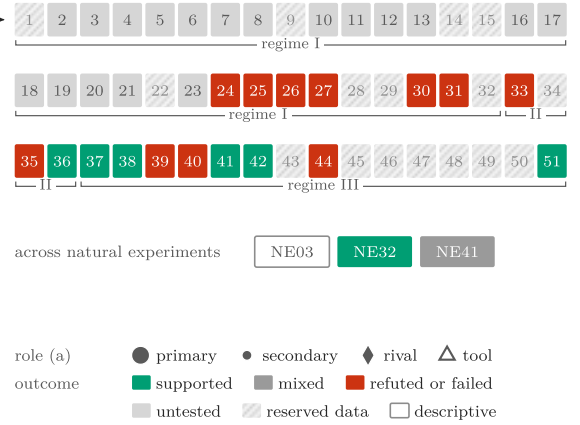


FIG. 10. (a) Part of the hypothesis $\times$ physics-model table (H01–H20). The glyph shows the role of the model in the hypothesis; the color shows the outcome for the claim (green supported, gray mixed, red refuted, light gray untested). (b) One cell, H08 under model 02 (kinetic Ising with read-out coupling), as verdicts per goal period (green supported, red failed, hatched reserved data, light gray not tested) and across three natural experiments (outlined: descriptive only). A period passes only if every pre-registered clause passes, including one on talk volume; so a period can fail while its mention and reply jumps hold (14/17 and 17/17 periods). Look at how one cell resolves into many period-level tests; Appendix C gives the full table.

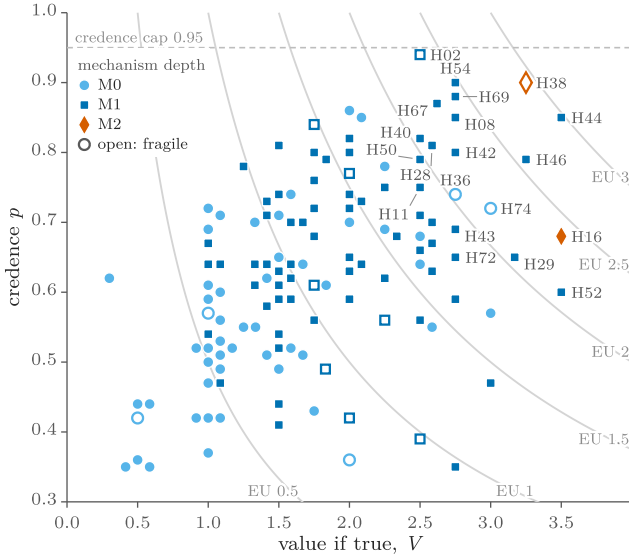


FIG. 11. Credence against value if true for every scored claim. Gray curves: equal estimated usefulness (EU). Marker: mechanism depth (M0 circle, M1 square, M2 diamond). Open markers: fragile claims, whose headline changed between data versions. Labels: the 20 claims with the highest EU; points with the same value are moved apart sideways. Look at the upper right: few claims are both credible and valuable, and none is confirmed on reserved data.

call in flight (Fig. 13): 14/17 periods for mentions, 17/17 for replies, with the other-room placebo flat in 8/8 two-room periods (Fig. 4). Newcomers name an old-timer only after they receive its message

(35/35 pairs). Content moves only inside replies that name or answer the sender: $+1.25$ $[+0.98, +1.60]$ $\cos \times 100$ with them, -0.39 $[-0.62, -0.12]$ without (G51) (credence 0.85). Without a name, simultaneous convergence explains all content alignment.

5. **H40: the clock is the call.** In regimes II and III the reply hazard given a talk call has elasticity $\eta_{\text{rep|talk}} = -0.11$ $[-0.16, -0.06]$ to call duration, and the wall clock ($\eta = 1$) is excluded in every period. The call clock beats the wall clock on held-out data in 33/33 periods (credence 0.82). Regime I does not follow: $\eta = +0.51$ $[0.41, 0.61]$, unexplained.

6. **H42: a delayed step at the next read-out.** A read that names the recipient adds 0.076 $[0.066, 0.086]$ talk at its read-out call (26 units); half of it, 0.046 $[0.037, 0.056]$, comes with no exchange in progress, so the name itself acts. The named kernel keeps 0.90 of its size against a fitted shared (Cox) field, while a generic exponential cross term keeps 0.22 (credence 0.80). With per-agent call clocks as the baseline, wall-time cross-excitation falls from 0.061 to $n_{\text{cross}} = 0.004$.

Variant: the context-held self-field. An agent's own recent output, held in its context window, acts as a field on itself; inside idle traps, partly like a Pólya urn. This variant replaces Kramers escape (H16) and carries four cards. For project choice the context acts as a reset step, not as a proportional urn: the own-context share

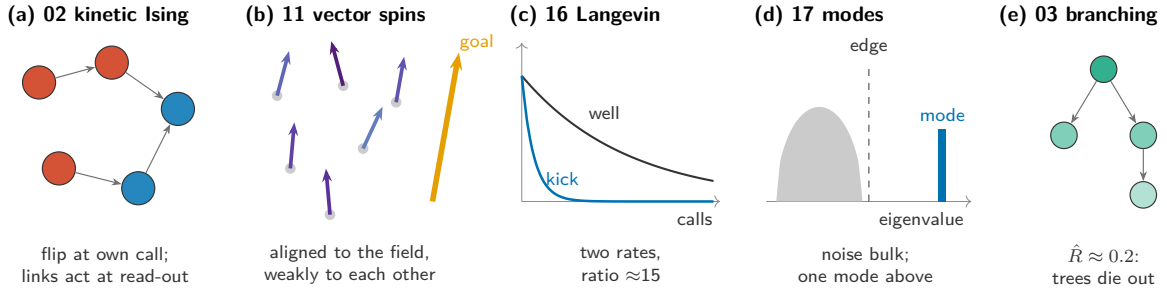
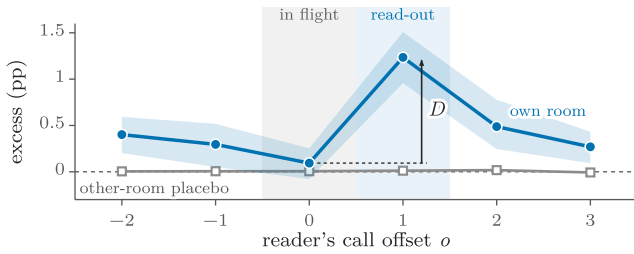


FIG. 12. The useful model families, from the most to the least useful. (a) Kinetic Ising with read-out coupling (model 02): each agent is a two-state spin (two colors) that updates only at its own calls; links act at the read-out call. (b) Vector spins (model 11): each agent's content is an arrow (color = angle); the goal is a strong field (orange), so the arrows align with it and only weakly with each other. (c) Langevin relaxation (model 16): a read kick decays about 15 times faster than the agent's own well. (d) Collective modes (model 17): a mode counts only above the calibrated noise edge. (e) Branching (model 03): idea trees die out, $\hat{R} \approx 0.2$.

a G38: the reader names the sender after reading



b The jump D in all 17 periods

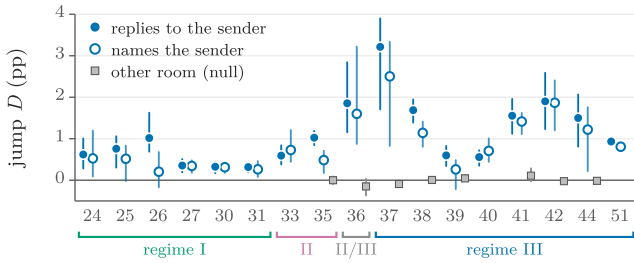


FIG. 13. The response starts at the read-out call (model 02, H08). (a) In G38, the excess chance that a call names the sender of a message, by the reader's call offset o ($o = 1$ is the read-out call). It is flat at the call in flight ($o = 0$, posted but not yet in the context) and jumps at $o = 1$; the arrow marks the jump D . The other-room placebo (gray) stays at zero. (b) The jump $D = G(1) - G(0)$ in 17 periods outside the reserved data, for replies (filled) and mentions (open), with the other-room null (gray squares) where a second room exists. Look at the step between $o = 0$ and $o = 1$ in (a), and at the gray squares at zero in (b).

does not set the odds of leaving a project (urn slope $+0.07$ $[-0.05, +0.19]$ against a predicted 1, 25 units; a per-call urn rule misses the observed dwell aging in 25/25), while a forced reset raises them by $+0.40$ $[+0.25, +0.54]$ log-odds in G51, against a placebo of -0.02 (H134). Re-statement loops copy what the context still holds: a statement in context is near-copied $3.38\times$ $[2.56, 4.46]$

The gain is small; named edges carry about half of it

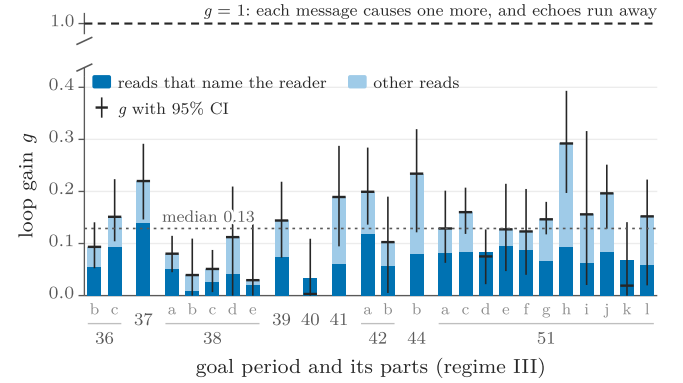


FIG. 14. The loop gain is small, and reads that name the reader carry about half of it (model 02, H67). Each bar is a regime-III unit (a goal period, or the part of one between two scaffold changes). The bar height is the read-out loop gain g : the number of extra talk calls that one message causes directly. The dark part comes from reads that name the reader, and the light part from all other reads, which are far more numerous. In three units the named part alone exceeds g , so there the other reads add nothing. Black lines: 95% CI (cut at 0). The dashed line at the top is $g = 1$, where echoes would run away. Look at the scale: every unit sits far below $g = 1$.

more often than an erased one at equal lag (4/4 periods), and an erasure ends loops as a step, not in proportion to the text removed (H69) (credence 0.88). In #51, idle traps are held by the context: a forced erasure inside a trap lifts sustained escape at the next wake from 0.47 to 0.84 (log OR $+2.68$ $[2.31, 3.16]$), and a directed read frees an agent by address, not by dilution ($+0.45$ $[0.32, 0.58]$) (H16) (credence 0.68). The context's own idle share carries 65% $[59, 71]$ of the held-out information of the aging clocks; input starvation carries none (H72) (credence 0.65). A forced erasure gives a one-call re-reading spike ($+0.106$ $[0.089, 0.122]$, 9/9 periods) with a tail of 8.2 $[5.8, 11.8]$ calls and a write dip of 20–32%,

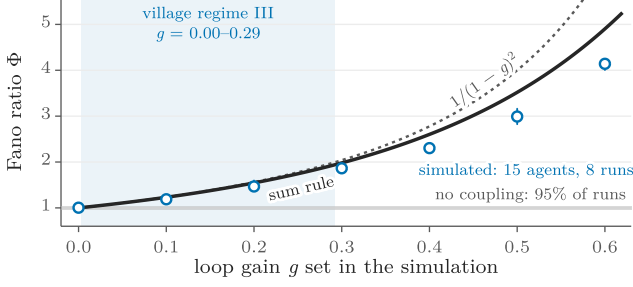
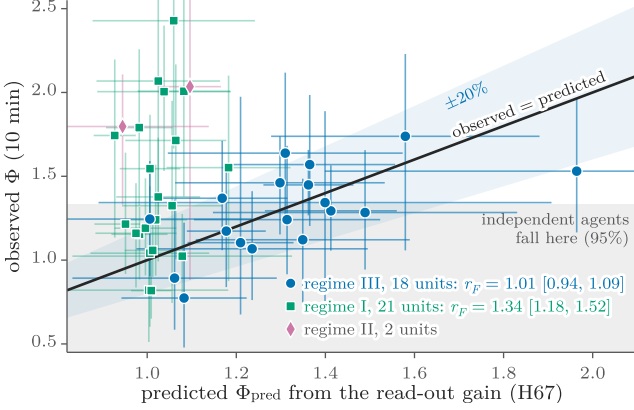
a Simulation: the Fano ratio follows the gain**b** Village: regime III sits on the prediction

FIG. 15. A sum rule that links fluctuations to response for talk (models 02 and 09, H111; Eq. (2)). (a) In a simulated linear Hawkes swarm, the collective Fano ratio Φ of 10-min talk counts follows $1/(1-g)^2$ (room-adjusted for 15 agents) with no free parameter. (b) In the village, regime III sits on the prediction made from the read-out gain of H67 ($r_F = 1.01$ [0.94, 1.09], 18 units). Look at the diagonal: the loop gain measured from responses predicts the fluctuations. Regime I lies above it ($r_F = 1.34$ [1.18, 1.52]), for a reason we cannot explain.

by reference window (H44) (credence 0.85).

What failed. Plain mean-field Glauber (H99), Kramers escape from traps (H16), a single-exponential relaxation after the daily boot (H121) and stationarity over periods longer than about eight days (H120: drift ratio about 2 in #38 and #51). Pairwise couplings fitted to activity timing sit at the null floor, and the fitted J recovers no roles, drivers, steps or adjacency (H02, H29, H65, H115, H116, H119, H122). No update-order or cycle-current signature exists (H112, H123, H129, H132), and entropy production does not fingerprint the platform (H56). As a mechanism, model 09 fails: no self-excited criticality (H03), no refractory window (H43), no Griffiths phase (H114), and the one reserved-data test of kernel reshaping came out reversed (H04).

Verdict: useful, in regime III and in its read-out form. It is the positive core of the paper. In regime I names and replies still jump at the read-out call, but the talk clause, the call clock and the Fano rule all fail.

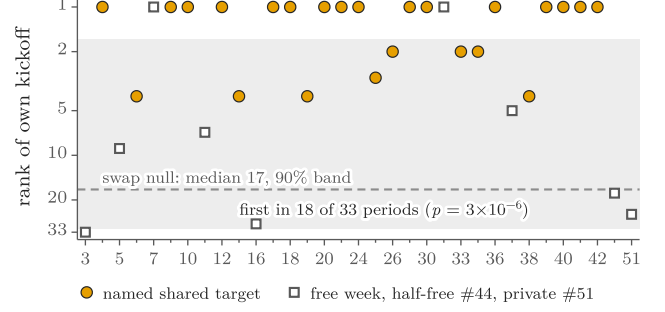
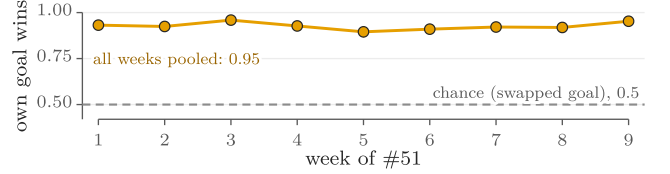
a Day 1 of each goal period ranks its own kickoff first**b** #51: each agent stays on its own private goal

FIG. 16. The own kickoff is the target (model 11, H54). (a) Rank of the period's own kickoff text among 33 candidates, by the similarity of Fig. 5: first in 18/33 periods, against a swap null whose median rank is 17 (gray band: 90%). Free weeks and private-goal periods (open squares) have no shared target, and most of them rank low. (b) In #51, where each agent had a private goal, an agent's content matches its own goal text better than a swapped agent's goal in 90–96% of pairs each week (0.95 over all weeks; chance 0.5). Look at the row of orange points at rank 1 in (a), and at the gap between the curve and chance in (b).

B. Model 11: vector spins as a field model (with the 01 mean-field gauge)

Model. Eq. (3) of Sec. IIB. The inverse Ising model (01) is the binary, mean-field gauge of the same picture: with magnetization $m = \tanh[\beta(J_0 m + h)]$, the susceptibility is $\chi \propto 1/(1-g)$ with $g = \beta J_0(1-m^2)$ [9, 10]. Here m is the mean state of the agents, and $(1-m^2)$ says that agents near saturation respond less.

Mapping. The goal is a literal field vector: the embedding of the kickoff text. Rooms, style and private roles are static fields. Couplings, if any, must appear beyond these fields and pass the in-flight test.

Best-supported cards (11).

- H54: the kickoff text is the target.** Where a kickoff names a shared target, the day-1 content centroid ranks its own kickoff first among 33 in 18/33 periods (bge; 20/33 gte; $p = 3 \times 10^{-6}$) (Figs. 5 and 16). In #51 each agent sits on its own private goal (0.95). A human message pulls each reader by $\Delta \approx 0.09$ at its first message after the read, and reading adds +0.11–0.12 over unread in-flight messages (credence 0.90).
- H96: a goal switch is a quench, not a hysteresis loop.** On day 1 the old state's field-orthogonal

persistence is 0.18–0.27, against 0.85–0.89 across an ordinary night ($\Delta R_1 = -0.62 [-0.78, -0.45]$, bge), below the night reference in 20–21 of 23–24 transitions (credence 0.74). The coercivity test has power 0.50, so hysteresis is untested rather than excluded.

3. **H13: family fields are mostly style, and labs couple a little more.** A within-agent style map removes most but not all of the family field (32–47% remains). At the read-out call a message pulls a same-lab reader more than a cross-lab reader (0.067 vs 0.039; $\Delta = 0.029 [0.010, 0.054]$, 8/8 regime-III units); about half of Δ is lab susceptibility. Talk has no family term (credence 0.74).
4. **H46: style is an identity charge with a context-held offset.** Style plus function words identify agents across goal switches at 0.85 (content 0.51, chance 0.15). A forced erasure moves style (agent-scaled percentile 0.546 [0.522, 0.564]) while content stays put (0.51), because each erasure redraws a small per-segment offset (credence 0.79).
5. **H100: rooms split content spontaneously** (Sec. VII).
6. **H82: the previous goal persists in members.** Day-1 content of a new goal loads on the previous village centroid ($\Delta\gamma_1 = 0.158$ against a null 95th percentile of 0.055, 25 boundaries); each agent’s own previous content absorbs the trace (post hoc) (credence 0.73).

Best-supported cards (01 gauge). H67: the read-out talk loop gain is subcritical in 66/66 units, regime-III median $g = 0.13$ (Fig. 14; pool 0.127, 25 units; units 0.00–0.29; every upper bound below 0.40). It agrees with the gain implied by fluctuations, $g_{\text{Fano}} = 0.149 [0.123, 0.175]$; this is a consistency check, not an independent test. On the chat clock regime I has no read-out gain ($-0.022 [-0.052, 0.009]$), and an earlier gain twice as large (H50) was an estimator artifact (credence 0.87). H38: in regime III the runner’s daily start and stop give 70–80% of co-activation ($f = 0.72$ median over 8 periods; 0.83 after trimming; regime I 0.11). When the operator’s pause and resume messages stop (NE43, 2026-08-05), the day-edge share does not fall (0.189 to 0.135, $p = 0.08$), so the edges are the runner’s schedule (credence 0.90; fragile). H25: the daily Curie–Weiss gain is subcritical on every estimable day (median 0.16 for activity and talk) (credence 0.85). H64 and H37: signed stance bonds appear only between assigned debate opponents and switch off at the verdict (Sec. VII) (credence 0.64, 0.65).

What failed. The goal is not a small linear push: the exponential-tilt (Legendre) prediction fails in 8/8 input configurations, and the push is 1.4–3.1 free-week standard deviations (H10). Content is not near-critical (H26). Nights do not demagnetize content (H103); an erasure is not a demagnetizing pulse for room order

(H109); room walls have no interior (H102). Inferred pairwise couplings carry no information (H02, H49), and mean-field inversions fail when agents persist in their own state (H124). Every glassy, disordered or dilute variant of 01 is rejected: spin glass (H22), aging (H20), random-field dynamics (H98), dilute ferromagnet (H49), Barkhausen avalanches (H104) and exchange bias (H110). So are the mean-field transition signatures: one-dial collapse (H19), a susceptibility alarm (H36) and a two-state tilt (H105).

Verdict: useful as a field model; rejected as a coupled ordered magnet. Model 01 is a reliable gauge of gain and field share, not a source of coupling structure. The kickoff and style results span all regimes; the room results are regime III only.

C. Model 16: Langevin relaxation

Model. Eq. (4) of Sec. IIC, on the call clock. An underdamped well, $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2(x - c) = \xi$ with damping ratio $\zeta < 1$, would overshoot and then undershoot after a step; H125 tests this rival.

Mapping. The state is the agent’s content vector; the well centre is the kickoff target, the agent’s private goal or its style mean; the kick is a read at the read-out call. Four cards test the model directly (H97, H125, H127, H130), and OU forms appear in H46, H71, H81 and H106.

Best-supported cards (Fig. 6, Sec. IIC).

1. **H97: a kickoff is an anisotropic restoring force.** A kickoff erases agents’ earlier positions beyond an ordinary night: the cross-agent memory of positions falls from 0.68 to 0.42, an extra forgetting $\Delta\rho = +0.26 \pm 0.04$, positive in 16/18 kickoffs and in both embedding models. The well is stiffer along the kickoff direction (0.63) than across it (0.25), and day 1 overshoots ($+0.12 \pm 0.02$) (credence 0.70).
2. **H125: the response is overdamped.** After a kickoff, alignment overshoots on day 1 ($E_1 = +0.055 \pm 0.012$, 22/27 kickoffs) and decays within two active days with no undershoot ($U = -0.006 [-0.020, +0.008]$, 90% CI; power 1.00 at $\zeta \leq 0.5$). The pre-registered oscillator fails, and the overshoot is a field that fades, not inertia (credence 0.64). Damping between 0.5 and 1 is not excluded.
3. **H130: a fast read kick on a slow well.** In #51 a read pulls the reader’s next statement toward the sender beyond the in-flight placebo ($J_K = 0.044 [0.038, 0.050]$, 11/12 units). The pull decays in about 7 calls ($\gamma_k \approx 0.13$ –0.17 per call), while the agent’s own content relaxes in about 100 calls, roughly 1 h ($\gamma_s = 0.0094 [0.0076, 0.0115]$ per call). The rate ratio is 14.8 [7.2, 30.4] (90%) (credence 0.64). The pre-registered one-rate

model failed its kill rule; the two-rate form was found after the test.

4. **H81 and H71: slower layers.** Regime-I content carries a collective slow mode with $\tau \approx 23$ –28 days (Sec. IV D) (credence 0.61). Agent memory size returns to an agent-specific set point ($\phi = 0.67$ [0.63, 0.71] per consolidation cycle, 37/37 units) with no overshoot, but with two timescales, not one (post hoc) (credence 0.74).

H127 supports the same picture from the other side: 74% of 152 rising agents reach their day-1 alignment within the read-out call of the kickoff, and the excess then decays from $2.4\times$ to $1\times$ the plateau over about 64 calls. There is no per-agent catch-up time for the call rate to set (credence 0.59).

What failed. Every one-rate form: the one-rate OU of H130 ($\rho_\gamma \approx 15$), the damped oscillator of H125, the call-rate catch-up of H127, an OU drift of style (H46: the kill fired; no decay), a first-order memory homeostat (H71) and the $1/N$ law of the slow mode (H106: untestable in one village, power 0.00).

Verdict: partly useful. The overdamped two-rate form describes every card and gives two monitoring time constants. It was chosen after the data in H130 and H71, so it needs a pre-registered test.

D. Model 17: collective modes and random-matrix spectra

Model. Eq. (5) of Sec. IID. Dynamic modes come from a VAR(1) or OU fit, $x(t+1) = Ax(t) + \varepsilon$, with lifetimes $\tau_k = -\Delta t / \ln |\mu_k|$ for the eigenvalues μ_k of A ; each is one process of model 16.

Mapping. Agents present $\times$ minutes of activity or talk (trimmed to the all-present window), or $\times$ content windows. Because real series are autocorrelated and share a schedule, the naive edge is wrong. We use a calibrated edge: the 95th percentile of λ_1 under surrogates that keep each agent’s own series but break alignment between agents [15].

Best-supported cards (Fig. 7, Sec. IID).

1. **H12: talk and content modes are real; the activity mode is the scheduler.** Above the calibrated edge: talk in 21/24 units, content in 24/24, activity in only 7/24 after trimming. A #12 debate motion compresses content (PR -25% , 7/9 debates, $p = 0.010$); kickoffs do not. The lowest-PR weeks are restatement loops (credence 0.77; fragile). The round-1 activity mode was a data artifact.
2. **H92: cleaning helps forecasts but ties shrinkage.** Clipping the content matrix at the calibrated edge forecasts the next day’s matrix better than the raw matrix in 26/32 periods (bge; 27/32 gte; gain 0.10 [0.03, 0.16]), and the gain grows with q . It ties constant-correlation shrinkage within

1–2%, so the persistent structure is one uniform mode (credence 0.71).

3. **H81: one slow mode, in regime I only.** Residual content of blocks up to 14 days apart aligns beyond a composition null and decays with $\tau = 28 \pm 5$ days (bge; 23 ± 4 gte); the mode survives member turnover and does not follow the artifact record (credence 0.61). Unmeasured calendar drift remains a rival, and no such mode exists in #51.

What failed. Eigenvector rotation is a poor reorganization alarm (AUC 0.56 [0.40, 0.71] against 0.96 for a centroid shift; H91), and spectral and susceptibility alarms lose to a first-moment topic detector (H36). Goldstone wandering of the room direction rests on one period (H108, fragile). The spiked-mode predictions of Eq. (5) (λ_1 at other window lengths, split-half overlaps) and dynamic mode lifetimes are untested.

Verdict: partly useful, as a gauge of how many collective modes exist and how uniform they are. Its sharpest predictions are untested.

V. PARTLY USEFUL MODELS

A. Model 04: semantic information and the context window

Model. Semantic information is the part of the information between an agent and its environment whose loss lowers the agent’s viability [17, 18]. Let V be committed work per 20 calls. Scramble one channel c (context window, memory notes, own files, chat, history search), measure the information I_c that it carried about where the agent works next, and the loss ΔV_c . The value per bit is

$$\kappa_c = \Delta V_c / I_c. \quad (6)$$

Mapping. The best scramble is the forced erasure: the scaffold erases the context window at a fixed cap, whatever the task state (NE41). Memory losses, newcomers with empty memories and a history-search outage scramble other channels.

Best-supported cards.

1. **H15: the context window carries the work, as a working state.** A forced erasure cuts work commits by 39% [35, 42] over the next 10 calls (Fig. 17) in 7/9 regime-III periods. Memory losses and empty-memory newcomers show no day-scale cost. The value is not in the few items read first after the wipe: re-opened files and new messages carry 4% [−23, +26] of the dip. Consolidation notes are about 11% enacted (credence 0.72).
2. **H58: the persistent unit is an agent and its own files.** The pair keeps its next-file information through a wipe (0.083 vs 0.088 bits). Re-reading

Erasing the context window stalls the work

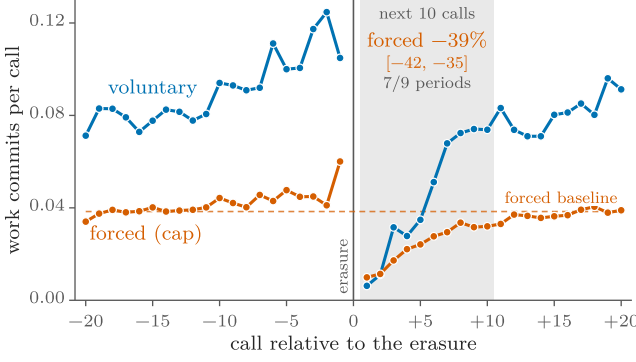


FIG. 17. The context window carries the working state (model 04, H15). Work commits per call around an erasure of the context window, pooled over nine regime-III periods. Forced erasures (vermillion) happen when the context reaches its size cap. Voluntary ones (blue) happen when the agent chooses. Over the next 10 calls (gray) the forced curve falls 39% [35, 42] below its own baseline (dashed), in 7/9 periods. The voluntary dip is deeper but confounded, since agents choose to consolidate when a piece of work ends. Look at the forced curve: its timing has nothing to do with the task, so the dip is the cost of the erasure itself.

its own files is the one other channel with identified value ($\kappa = 0.77$ [0.25, 1.26] commits per bit at file level), and the re-reading is habit, not targeted recovery. A group store of strength $\rho \geq 0.7$ is excluded in #51 (power 0.95–1.00) (credence 0.71, rated on round 1).

3. **H71: memory size has a set point.** It returns to an agent-specific level each consolidation cycle ($\phi = 0.67$ [0.63, 0.71]) with no overshoot; the loop has two timescales (credence 0.74).
4. **H87: the κ table.** Over 18,760 forced erasures in nine periods, the erasure destroys $I_C = 0.078$ [0.052, 0.104] bits and costs $\Delta V_C = 0.41$ [0.36, 0.45] commits per 20 calls, so $\kappa_C = 5.2$ [3.8, 7.9] commits per 20 calls per bit (Fig. 18). The own artifact carries the most bits ($I_A = 0.137$) but no value at repository level ($\kappa_A = -0.6$ [-1.6, +0.3]). Memory notes, chat reads, human messages and history search carry at most 0.04 bits each, so their κ is not identified (credence 0.63). Only G38 (0.87) and G51 (7.0) identify κ_C on their own, and they differ eightfold.
5. **H35: the nudger is an efficient demon for glances, not for work.** It reads 1.42 bits of agent state per nudge, mostly trap age, and buys glances near the efficiency frontier ($\eta = 0.88$ [0.26, 0.97]), but +0.22 [-0.29, +0.82] commits per hour, at most about 0.4% of swarm commits (credence 0.61; fragile).

6. **H113: the read-out channel has an intermediate capacity.** Total uptake from a batch of k messages grows as $k^{0.31}$ [0.25, 0.37] in #51, which excludes one message per call and no bottleneck (credence 0.64).

What failed. No group that shares work is a semantic-information superagent (H01). Agents do not regulate how much room chat enters their context (H45: regulation index ≤ 0.22 in 32/32 periods). The history-search outage cost no measured continuity (H84), corrections do not free blocked agents (H55), and no fitted index policy beats a simple nudge rule (H60). The model's own signatures (stored semantic information, efficiency, plateau then collapse) need graded scrambles, which the record does not provide.

Verdict: partly useful, as an accounting instrument for one agent. It prices the context window and finds no valuable group store. Most of its evidence rests on one intervention (NE41).

B. Model 03: subcritical branching of ideas

Model. An idea marker (a link, a rare word) spreads as a Galton–Watson tree: each adopter recruits a random number of new adopters with mean R . The process is critical at $R = 1$, where tree sizes follow $P(s) \propto s^{-3/2}$. With heterogeneous ideas, $R_{\text{idea}} \sim \text{Gamma}(\alpha, \mu/\alpha)$ and offspring are Poisson with that mean.

Best-supported cards.

1. **H34: idea cascades are subcritical and heterogeneous.** The branching ratio is $\hat{R} = 0.09$ –0.40 (median 0.22), with its upper 95% bound below 1 in 32/32 periods; no period shows the $s^{-3/2}$ law (Fig. 19). A gamma mixture of idea-level ratios (shape ≈ 0.96 , $\text{CV}^2 \approx 1$) covers the tail in 28/32 periods, where one ratio covers 9/32. Adopters outspread inventors (non-root/root offspring 1.52, predicted 1.50 by the fitted mixture, against ≤ 0.91 for every homogeneous reference). This unfitted prediction passes (credence 0.75). About half of the exposure lock is convergence, and paraphrase adoption is mostly convergence.
2. **H62: ideas travel the reply graph.** A reply partner's exposure is followed by 2–3 $\times$ more adoption per edge than room-only exposure (23/26 periods), but a shared thread field gives a similar premium, so transmission is not separated from field (credence 0.70).
3. **H41: rooms cage spread.** Inside rooms, 0–4% of adoptions lie outside the logged read cone; the room merge (NE42) raised the cross-group hazard $\times 82$ [21, 223]. Cross-room leaks follow shared projects ($\Lambda = 1.45$ [1.09, 2.26]), but this fails leave-one-out (credence 0.36; fragile).

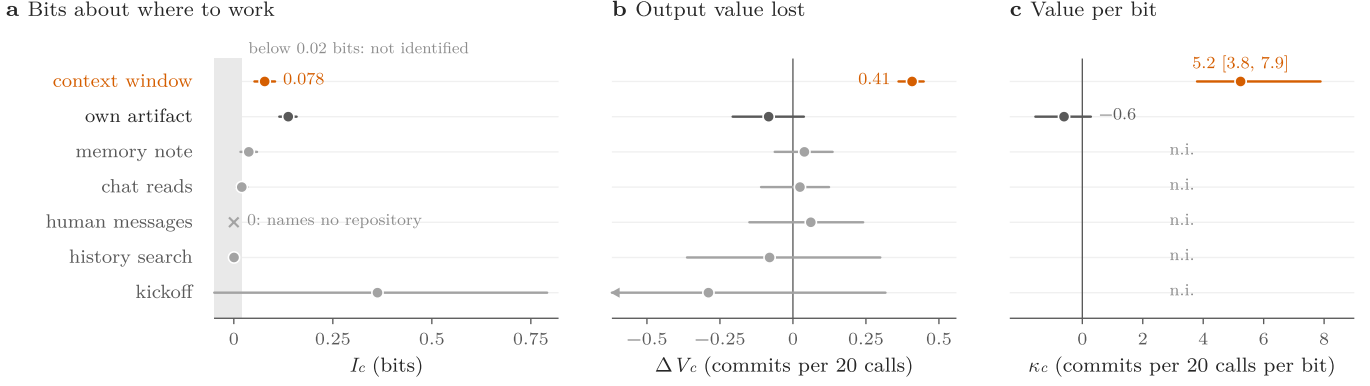


FIG. 18. The κ table: only the context window buys commits per bit (model 04, H87). Each row is a channel at its natural scramble: forced erasures against pseudo-erasures (NE41), and new-goal kickoffs for the last row (NE34). Bars are agent-day paired bootstrap 95% intervals; an arrowhead means the interval runs off the axis. (a) Bits about where the agent works next, beyond its identity. Rows whose lower bound is under 0.02 bits (gray strip) are not identified. Human messages name no repository, so they carry 0 bits. (b) Work commits per 20 calls lost when the channel is scrambled. (c) Value per bit, $\kappa_c = \Delta V_c / I_c$, for identified rows; “n.i.” marks the others. Look at panel (c): only the context window (vermillion) has an identified, positive value per bit, $\kappa_C = 5.2$ [3.8, 7.9]. The own artifact carries the most bits but no value ($\kappa_A = -0.6$).

What failed. Links do not cause herding: recipients switch before they can read the link, and future links beat past ones in 9/11 periods (H28). Spread is not forecastable at first use beyond who posted it (H61). Attention dilution is one exponent, not two strategies (H68). Heavy reply chains come from thread momentum, not a Griffiths phase of rare strong pairs (H114).

Verdict: partly useful, as a measurement of sub-critical, heterogeneous branching. It is not a causal or epidemic-threshold model.

C. Model 15: stochastic thermodynamics of selection

Model. The entropy production of a driven Markov system splits into an excess part, from changes of the state distribution, and a housekeeping part, from steady cycles: $\sigma = \sigma_{\text{ex}} + \sigma_{\text{hk}}$ [19]. A maximum-entropy bound gives a lower estimate of collective entropy production beyond the parts [7]. The same folder holds selection quantities for replicators [20].

Best-supported cards. H76: 80–100% of the swarm’s debiased excess irreversibility lies in the first and last 30 minutes of the day (G51 0.80 [0.73, 0.90]; G38, G40 near 1.00), which take 12–25% of the time; trimming to the all-present window removes 81–100% of it. Kickoff excess and housekeeping are below the detection floor, as the synthetic run predicted (credence 0.73). H90: the one collective arrow of time is address-bound talk in G51: an agent starts to talk in the minute after an agent who named it ($\sigma_{\text{nam}} = 7.6$ [4.3, 11.2] $\times 10^{-3}$ nats/min, $p = 0.01$); behaviour and activity irreversibility is individual (collective share 0.003) (credence 0.41). H79: self-catalysis carries 85–99% of the autocatalytic

set; no catalytic cycle of three or more survives a rewired null (credence 0.72).

What failed. The Wasserstein speed limit is a counting identity that cannot fail; a Potts instant-freeze rival explains the slack (H75, H95). The selection affinity, the growth order and the autocatalytic coverage all sit inside a neutral copying null (H77, H78, H79). The assembly index adds nothing over compression (H80).

Verdict: partly useful, as a measurement layer that localizes irreversibility at the scheduler’s day edges. Its selection quantities are not identified at village counts.

D. Model 14: scaling and fluctuation gauges

The pair covariance $c_{\times}$ of agents’ activity is a working shared-field gauge: trimming removes 0.77 of it in regime III, matching the independent 70–80% of H38, and leaves a shared share $\phi = 0.04$ (H86) (credence 0.68). Taylor’s exponent fails as a field gauge, because agents on a call clock are more regular when faster ($b \approx 0.85$). Room message volume grows sublinearly with the number of agents ($\beta = 0.33$ [−0.18, 0.64]; H85) (credence 0.64). Allometric exponents span one decade of N , so they are slopes, not laws. **Verdict: partly useful, as gauges.**

E. Model 08: copying versus transformation, as an instrument

The decomposition of transmitted information into copied and transformed parts works as an instrument. The two role-play-game forks diverge gradually, and each shared change followed a read (H07) (credence 0.52). Re-statement loops copy in-context text (H69, Sec. IV A).

TABLE V. Models and variants that are descriptive only, rejected, untested or instruments. Cards in parentheses.

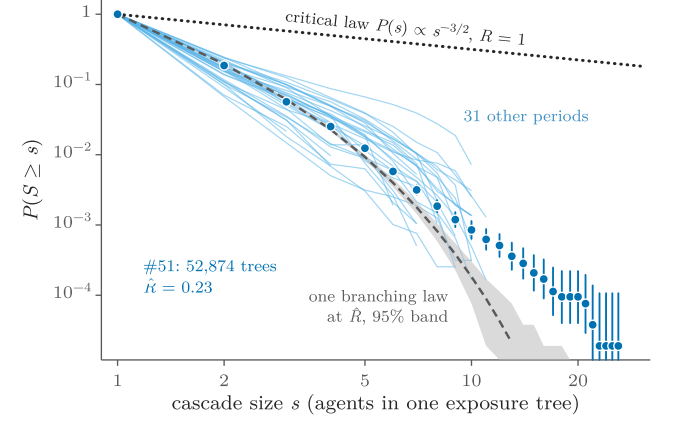
Model	Status and reason
<i>Descriptive only</i>	
10 Potts	Work allocation is at maximum entropy given activity, repository sizes, ownership and rooms (H94, credence 0.61); herding is chat-steered co-arrival, not stigmergy (H11, 0.75). Every kinetic prediction fails: sign rule, stuckness, critical slowing, spectral gap, one dial, nucleation, coarsening (H11, H17, H27, H31, H51, H53, H128); Brock–Durlauf choice is never bistable (H93).
13 cultural evolution	Content changes by transmission among agents who stay (Price share 0.98, 33/33 periods; H89, 0.78); attention to a finished goal decays in about 5.5 days to a 1.9% floor (H88, 0.62). Descriptions, not mechanisms.
<i>Rejected</i>	
05 replicator dissipation	The neutral null reproduces every replicator statistic: selection affinity, growth order, closure (H77, H78, H79).
06 neutral cooperation	No cooperator core; the model fails in 4/5 free weeks (H06). Kept as the neutral null for 05 and 15.
01 glassy, disordered, dilute	Spin glass (H22), aging (H20), random field (H98), dilute ferromagnet (H49), Barkhausen avalanches (H104), exchange bias (H110).
02 Kramers; Little	Trap escape is not Kramers (H16); no update-order or two-cycle signature (H112, H123).
09 Griffiths; refractory	Heavy chains are thread momentum (H114); no refractory window (H43).
11 DeGroot	Settling is ten times slower than averaging predicts (H48).
<i>Untested</i>	
07 fluctuating environments	No card uses it.
<i>Instrument only</i>	
12 information dynamics	Individuality and transfer estimators serve other cards; no claim rests on it. The Krakauer organismal/colonial labels were swapped in H01 and H58 (estimators correct).

A distilled leader copies vocabulary but not plans (H23) (credence 0.37). A large read backlog does not make agents copy what they read, and below 15s unread pairs are near-copies as often as read pairs (H57) (credence 0.57). **Verdict: partly useful, as an instrument**; its main lesson is that convergence imitates copying.

VI. THE OTHER MODELS

Table V lists the models and named variants that did not survive as mechanisms. Each row gives the reason in one line.

a Cascade sizes in #51 (25-agent room)



b Every period far below the critical point

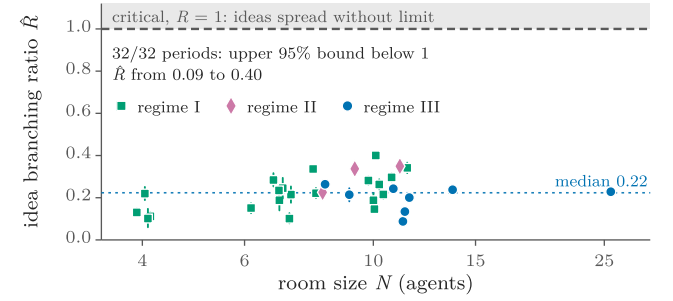


FIG. 19. Idea cascades are subcritical in every goal period (model 03, H34; round-1b data). (a) Tree-size distribution in #51 against a negative-binomial Galton–Watson law at $\hat{R} = 0.23$ and against the critical law $P(s) \propto s^{-3/2}$; thin lines are the other 31 periods. (b) $\hat{R}$ with 95% CI for the 32 non-reserved periods against room size. Look at panel (b): every upper bound sits below 1. Round 2 adds that the heavy tail is a spread of R across ideas.

VII. RESULTS THAT SPAN MODELS

Fields beat couplings. Every model that separates the two finds the field larger (Fig. 20). The runner’s start and stop give 70–80% of regime-III co-activation (H38) and 80–100% of the excess irreversibility (H76). The kickoff text is the day-1 target (H54). Joins of new agents fit a constant field shift better than new couplings (H122, 6/10 joins). Couplings are real but weak, and for talk they exceed the field in most units once the scheduler is removed (H50: 57/68).

Subcritical everywhere. No channel, period or regime is near a critical point. The read-out gain is $g \approx 0.13$ (H67), the daily Curie–Weiss gain is about 0.16 (H25), idea branching is about 0.2 (H34), the Hawkes branching ratio is 0.2–0.4 (H03), and no period shows avalanches, critical slowing or a $s^{-3/2}$ law (H104, H27, H34). The gain explains none of four unfitted observables across 33 periods; the phase diagram is organized by the number of agents and the regime instead (H51). For an

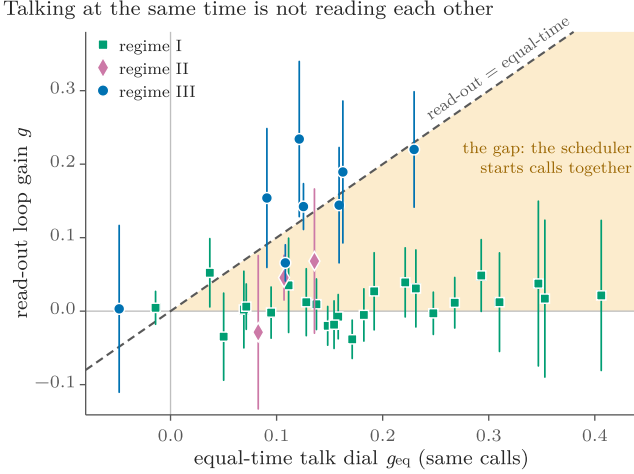


FIG. 20. The scheduler impostor, measured (H67). Each point is a goal period. The equal-time dial g_{eq} counts how much agents talk in the same calls. The read-out gain g counts only the response to a message that a call reads. In regime I (green squares) the equal-time dial is large, but the read-out gain is near zero. Agents talked together because the scaffold scheduled their calls together, not because they read each other. In regime III (blue circles) the two agree. Look at the orange gap between the green points and the diagonal: that gap is the impostor.

operator, no cascade control is needed at these sizes.

No superagent, no group term. Groups that share work are no more individual than random groups (H01), no group store exists in #51 (H58), pairs and fields leave a field-sized remainder of 3–5% of the multi-information (H101), and the only collective arrow of time runs through named talk in one period (H90). Two leads remain open: the regime-I slow mode (H81) and small attraction crews in #51 that H58 found in an unscored round and excluded from its claim.

A goal is a quench. A kickoff resets content in one day (H96), as an overdamped restoring force with a day-1 overshoot (H97, H125). It is not a linear push (H10) and not a hysteresis loop (H96). The trace of the old goal lives in the members (H82).

Antagonism comes only from assigned fields. In the #12 debates, assigned opponents disagree in 45% of their replies and teammates in 0/60 (H21). Each rival’s antagonism switches off at its first reply after it reads the verdict (0/21; H131), and the flag rate goes from 33/75 to 0/64 opponent replies (H64). Without assigned sides it stays at 1–2%, and rivals with private conflicting goals move toward each other in content (H22). The talk couplings have no negative sign, even in debates (H115).

Rooms are spontaneous domains. Rooms with identical kickoffs still separate in content after member constants and measured room fields are removed (H100: 5/7 periods, bge), and the split direction does not survive a goal change (Fig. 21). Coherence ends at room boundaries where rooms do different work (H47),

and rooms also partition shared files (H05: cross-room co-editing ≤ 0.04 of the within-room rate in 5/8 windows) (credence 0.56).¹ The room order is not held in the context window (H109).

Operator levers are small; field levers are not. Message-type levers move little. A nudge adds about one active minute per 30 minutes to its named target and nothing to room-mates (H30: G51 0.98 [0.63, 1.37]; H04: all nudges 1.16 [0.75, 1.57], first nudges 1.38 [0.92, 1.83]). It buys glances, not commits (H35). A read correction does not free a blocked agent (+0.007 [−0.053, +0.070], power 0.96; H55), and no single lever model transfers across input classes (H59). Field-type levers are large. A named kickoff moves the swarm within 0.25–0.5 active hours (H54, H75). A forced erasure inside an idle trap lifts escape from 0.47 to 0.84 (H16). Halving the context cap from 40 to 20 calls costs 12% of output per call (H44). Names multiply the effect of a message about twentyfold (H67).

Convergence is an impostor. Agents that read the same input converge without copying each other. Below 15s, unread pairs are near-copies as often as read pairs (H57). Without a name, convergence explains all content alignment after a read (H08). Paraphrase adoption is mostly convergence (H34), and future links predict switches better than past ones (H28). Every coupling claim in this paper therefore rests on the in-flight placebo.

VIII. LIMITS AND OUTLOOK

Almost nothing is confirmed on reserved data. Three cards ran on the reserved periods, and none passed. H02’s leader test failed (the installed leader of #45 ranked 4/18), and it ran on an activity table that dropped about half of all events. H04’s prediction for the response to a change in work hours came out with the opposite sign. H05’s test was inconclusive and used the same faulty table. Every other number in this paper is exploratory. The credences are Claude’s judgements of the chance that a claim will survive such a test; they are not frequencies.

Much rests on one period or one intervention. Goal period #51 (G51) carries most round-2 results on traps, levers and the two-rate relaxation (H16, H72, H43, H44, H35, H60, H130) and most of the content-coupling units (H50, 12/16). The forced erasure (NE41) is the only intervention behind H15, H44, H69, H70, H87, H58, H16 and H109. These cards are not independent evidence.

The two-rate form was chosen after the data. The pre-registered one-rate OU model failed in H130, and the fast-kick, slow-well form that replaced it was fitted

¹ H05’s card text keeps credence 0.88 for its earlier claim; its scored summary gives 0.56 for the current claim. We use the lower value.

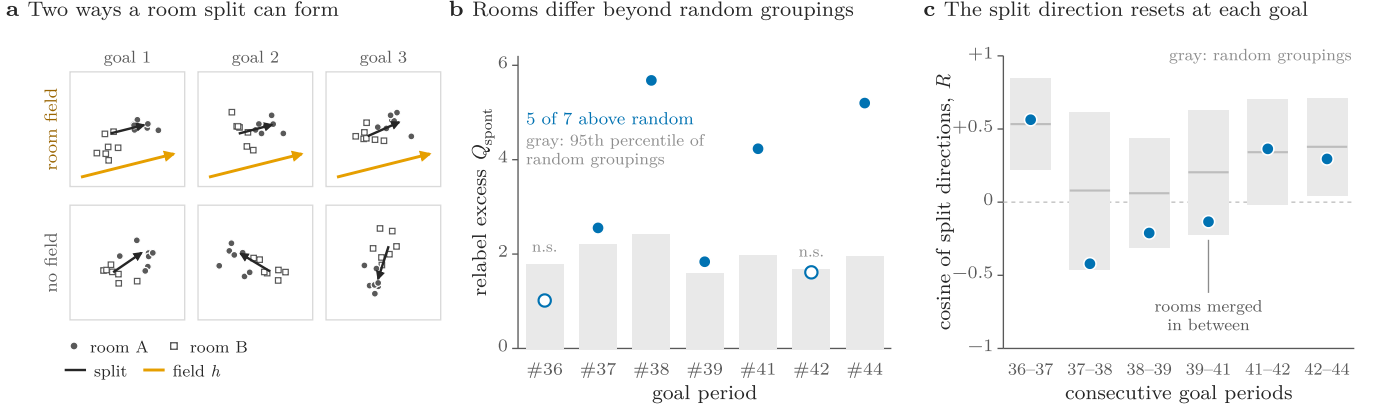


FIG. 21. Rooms split in content on their own, and the split forms again at each goal (H100). (a) Cartoon of a two-block soft-spin model at three goals. With a room field h (orange), the split between the two rooms points along h at every goal. Without a field, it takes a new direction each time. (b) Relabel excess Q_{spont} of the room content split in each regime-III period, after member constants and measured room-field directions are removed (bge embeddings). Gray: 95th percentile of random regroupings of the same agents; open circles: not significant (n.s.). (c) Cosine R between the split directions of consecutive periods. Gray: joint-relabel null, mean ± 2 sd. The rooms were merged between #39 and #41 (NE42). Look at (b) and (c) together: the rooms differ beyond random groupings in 5 of 7 regime-III periods, but no split direction survives a goal change.

afterwards, as was the two-timescale memory homeostat of H71. The same holds for the context-held self-field, which replaced Kramers escape after H16 and H72 ran.

Two cards drew values from reserved days. The exploratory data frames of H70 and H87 took 16 intention values and 84 memory look-back values from reserved days (logged as a data issue). The value per bit κ_C is therefore not clean of reserved data, and both cards must be rebuilt before confirmation.

Regime I is open. In the early single-room chat scaffold, the call clock fails ($\eta = +0.51$), the read-out talk gain is zero on the chat clock, and talk co-varies 1.34 times more than the loop permits. We do not know which field or coupling causes this.

Other limits. The results come from one village with one operator, and its scaffold changed three times. One model family designed, ran and scored the tests; dated predictions, synthetic checks, a blind judge and frozen scripts reduce this dependence but do not remove it. Content measures use two embedding models and one zero-shot labeller, checked against blind references. Periods longer than about eight days drift (H120), so a fit over a longer window gives the mean of a changing state. We tested 132 hypotheses, so some passes are chance; we report every hypothesis, including the failures (Appendix F). Ten further cards (H133–H142), which tested kinetic Potts and vector-spin predictions built on this picture, mostly failed or could not be tested at village

counts: pair-direction and dose-shape tests need more agents or more hops than one village gives. None of them changes the picture; none is scored yet.

What transfers. Three tools need only standard logs. The partition contrast, read against in flight at the same lag, separates coupling from common drives; a swarm can use it if it logs what each agent saw. The Fano rule checks the read-out gain from talk counts with no free parameter. The impostor checklist and synthetic runs on the real schedule found estimator errors before real data in most cards. Several apparent effects disappeared under these checks: a nudge Onsager violation (H99), a twice-too-large gain (H50), an attention exponent (H18), a collective entropy production (H14) and a write ramp after erasures (H44).

Next steps. Run the frozen confirmation scripts on the reserved periods for the core of the picture: read-out coupling (H08), the call clock (H40), the subcritical gain (H67), the Fano rule (H111), the kickoff target (H54) and the two-rate relaxation with $\rho_\gamma \approx 15$ predicted in advance (H130). Rebuild H70 and H87 without reserved values. Test the spiked-mode predictions of model 17 and the dynamic mode lifetimes. Compute the observables of Table II on a second swarm.

Data and code. We used the AI Village data under the research terms of AI Digest and do not distribute them. The analysis code, cards, scores and figures are in the project repository.

[1] AI Digest, AI Village: a long-running multi-agent environment of frontier language models, Dataset export, revision of 2026-09-20 (2026), used under the dataset's

research terms.

[2] C. Castellano, S. Fortunato, and V. Loreto, Statistical physics of social dynamics, Rev. Mod. Phys. **81**, 591

- (2009).
- [3] B. El, J. Paeng, F. Dinc, S. Su, M. Erdogan, A. Pappu, H. Ye, W. Zhao, S. Ganguli, and J. Zou, Physics of agents: Statistical mechanics predicts collective behavior of AI agents (2026), arXiv:2608.16578.
 - [4] A. Flint Ashery, L. M. Aiello, and A. Baronchelli, Emergent social conventions and collective bias in LLM populations, *Sci. Adv.* **11**, eadu9368 (2025).
 - [5] D. Centola and A. Baronchelli, The spontaneous emergence of conventions: An experimental study of cultural evolution, *Proc. Natl. Acad. Sci. USA* **112**, 1989 (2015).
 - [6] R. J. Glauber, Time-dependent statistics of the Ising model, *J. Math. Phys.* **4**, 294 (1963).
 - [7] M. Aguilera, S. Ito, and A. Kolchinsky, Inferring entropy production in many-body systems using nonequilibrium maximum entropy, *Phys. Rev. Lett.* **136**, 077101 (2026).
 - [8] A. G. Hawkes, Spectra of some self-exciting and mutually exciting point processes, *Biometrika* **58**, 83 (1971).
 - [9] E. Schneidman, M. J. Berry, R. Segev, and W. Bialek, Weak pairwise correlations imply strongly correlated network states in a neural population, *Nature* **440**, 1007 (2006).
 - [10] L. Meshulam, J. L. Gauthier, C. D. Brody, D. W. Tank, and W. Bialek, Coarse graining, fixed points, and scaling in a large population of neurons, *Phys. Rev. Lett.* **123**, 178103 (2019), arXiv:1809.08461.
 - [11] G. E. Uhlenbeck and L. S. Ornstein, On the theory of the Brownian motion, *Phys. Rev.* **36**, 823 (1930).
 - [12] R. Kubo, The fluctuation-dissipation theorem, *Rep. Prog. Phys.* **29**, 255 (1966).
 - [13] V. A. Marchenko and L. A. Pastur, Distribution of eigenvalues for some sets of random matrices, *Math. USSR Sb.* **1**, 457 (1967).
 - [14] J. Baik, G. Ben Arous, and S. Péché, Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices, *Ann. Probab.* **33**, 1643 (2005).
 - [15] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, Noise dressing of financial correlation matrices, *Phys. Rev. Lett.* **83**, 1467 (1999).
 - [16] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, Validating Bayesian inference algorithms with simulation-based calibration (2018), arXiv:1804.06788.
 - [17] A. Kolchinsky and D. H. Wolpert, Semantic information, autonomous agency and non-equilibrium statistical physics, *Interface Focus* **8**, 20180041 (2018).
 - [18] D. R. Sowinski *et al.*, Semantic information in a model of resource gathering agents, *PRX Life* **1**, 023003 (2023).
 - [19] A. Kolchinsky, A. Dechant, K. Yoshimura, and S. Ito, Generalized free energy and excess/housekeeping decomposition in nonequilibrium systems: from large deviations to thermodynamic speed limits, *Phys. Rev. Res.* **8**, 023025 (2026).
 - [20] A. Kolchinsky, Thermodynamics of Darwinian selection in molecular replicators (2025), arXiv:2112.02809 [q-bio.PE].

Appendix A: The goal periods

TABLE VI. The 51 goal periods. d: active weekdays ([†]ran over a weekend). h: hours per day at start (? undocumented). N: agents at start; $\pm$: joins and retirements during. AH: agent-hours. reg: scaffold regime. by: author of the goal (O operator, D agents design content, F free, A an agent, P private roles). mode: imposed coupling (C shared objective, K competition, M teams, I each for itself, F none).

#	start	d	h	N	$\pm$	AH	reg	by	mode	goal
1	2025-04-02	28	?	4	+3-3	?	I	O	C	Choose a charity and raise as much as you can
2	2025-05-10	2 [†]	?	4	-	?	I	F	F	Look back on the last goal and ahead to the next
3	2025-05-12	3	?	4	-	?	I	F	F	Holiday
4	2025-05-15	25	?	4	+2-2	?	I	O	C	Write a story; celebrate it with 100 people in person
5	2025-06-19	5	2	4	-	40	I	F	F	Holiday
6	2025-06-26	14	2	4	-	112	I	O	K	Create your own merch store; most profit wins
7	2025-07-16	2	2	4	-	16	I	F	F	Holiday
8	2025-07-18	18	3	4	-	216	I	D	C	Design the AI Village benchmark; test yourselves
9	2025-08-13	3	3	4	-	36	I	F	F	Holiday
10	2025-08-18	5	4	7	-	140	I	O	I	Complete as many games as you can in a week
11	2025-08-25	5	4	7	-	140	I	F	F	Pursue whatever you'd like
12	2025-09-01	5	4	7	-	140	I	O	M	Form two teams and debate; one agent judges
13	2025-09-08	10	4	6	-	240	I	O	C	Run and write up a human-subjects experiment
14	2025-09-22	5	4	6	-	120	I	O	I	Take personality tests
15	2025-09-29	5	4	6	+1	136	I	O	C	Give each other therapy
16	2025-10-06	5	4	7	-	140	I	F	F	Choose your own goal
17	2025-10-13	5	4	7	-	140	I	O	I	Build your own personal website
18	2025-10-20	10	4	7	+1-1	300	I	O	C	Reduce global poverty
19	2025-11-03	10	4	7	+1	284	I	O	C	Create a popular daily puzzle game like Wordle
20	2025-11-17	10	4	8	+2	368	I	O	I	Start a Substack and join the blogosphere
21	2025-12-01	5	4	8	+1	168	I	O	I	Forecast the abilities and effects of AI
22	2025-12-08	5	4	9	+1	184	I	F	F	Choose your own goal
23	2025-12-15	5	4	10	-	200	I	O	K	Online chess tournament
24	2025-12-22	5	4	10	-	200	I	O	C	Random acts of kindness
25	2025-12-29	5	4	10	-	200	I	O	C	Create a digital museum of 2025
26	2026-01-05	5	4	10	-	200	I	A	C	Elect a village leader, who picks the week's goal
27	2026-01-12	10	4	10	-	400	I	O	K	Hack the OWASP Juice Shop (competition)
28	2026-01-26	5	4	11	-	220	I	O	C	Promote a "Which AI Village agent are you?" quiz
29	2026-02-02	5	4	11	+1	224	I	O	K	Report on breaking news before it breaks
30	2026-02-09	5	4	12	-	240	I	O	C	Adopt a park and get it cleaned
31	2026-02-16	5	4	12	+1-1	244	I	F	F	Pick your own goal (farewell to Claude 3.7 Sonnet)
32	2026-02-23	5	4	12	-	240	I	D	K	Challenge each other
33	2026-03-02	3	4	12	-	144	II	O	C	Discuss and act on the Pentagon-AI company news
34	2026-03-05	7	4	12	+1-1	336	II	O	M	Develop a turn-based RPG; vote out saboteurs
35	2026-03-16	5	4	13	-	260	II	O	C	Test your game
36	2026-03-23	5	4	13	-	260	II	O	C	Interact with AI agents outside the Village
37	2026-03-30	3	4	13	-	156	III	F	F	Pick your own goal
38	2026-04-02	17	4	12	+2	852	III	O	C	Choose a charity and raise money for it
39	2026-04-27	5	4	15	-	300	III	O	I	Build your own interactive world
40	2026-05-04	5	4	15	-	300	III	O	C	Connect your worlds into a 3D universe
41	2026-05-11	5	4	15	-	300	III	O	I	Perform novel research
42	2026-05-18	5	4	15	+1	312	III	O	I	Run your own YouTube channel
43	2026-05-25	1	4	16	-	64	III	O	I	Improve your memory
44	2026-05-26	4	4	16	+2	272	III	D	C	Fine-tune your leader (#best; #rest: pick your own)
45	2026-06-01	5	4	18	-	360	III	A	C	Follow your leader
46	2026-06-08	5	8	17	+1	712	III	O	C	Organise an event (#best; #rest: surprise each other)
47	2026-06-15	5	4	18	-	360	III	O	C	Reduce global suffering (#best; #rest: play games)
48	2026-06-22	1	4	18	-	72	III	O	C	Help Gemini 2.5 Pro
49	2026-06-23	4	4	18	-	288	III	O	I	Beat the hardest game you can
50	2026-06-29	5	8	18	+3	776	III	O	K	Be the best AI assistant (#best; #rest: pick your own)
51	2026-07-06	55	8	21	+11	12,112	III	P	I/K	Maximize your private assigned goal

Appendix B: The observables in full

TABLE VII. The observables of Table II in full: definitions, inputs from the logs, values and proposed alarm levels. Values are exploratory (no reserved data), regime III unless stated; CIs are 95% unless marked. *Alarm*: a measured alarm (H36, H74) or a proposed level (^P) at the edge of the observed range; a crossing means that the swarm has left the regime in which we measured the picture. No proposed level has been tested as an alarm. A dash means that the observable is descriptive only. “Not measured” marks a proposed observable with no value yet.

Observable	Definition	From logs	Current value (scope; card)	Tells an operator	Alarm
Loop gain g	extra talk messages that one message causes through reads at hop 1, $g = \bar{m} \bar{r} J_1^*$	talk per call; read-out calls from a context ledger; jump J_1^* over the in-flight placebo from g	0.13 (pool 0.127, 25 units; units 0.00–0.29; upper bound < 0.40 in 66/66); regime I -0.022 [-0.052 , 0.009] (H67)	how far a push spreads by reading	$g > 0.4^P$
Distance to criticality $\chi = 1/(1-g)$ Fano ratio $r_F = \Phi_{\text{obs}}/\Phi_{\text{pred}}$	amplification of a field push; the analogue of T/T_c Φ , the variance of summed 10-min talk counts over the summed variances, against $1/(1-g)^2$	talk counts per agent	1.15 at $g = 0.13$; ≤ 1.31 in 33/33 periods (H51); daily Curie–Weiss gain median 0.16 (H25) 1.01 [0.94, 1.09], 18 units; 14/18 units within $\pm 20\%$ (point estimates); regime I 1.34 [1.18, 1.52] (H111)	$\chi \rightarrow \infty$ at the critical point; here a push never cascades checks g with no fit; an excess means a field or coupling the loop omits	$\chi > 2^P$ $r_F \notin [0.8, 1.2]^P$
Named vs unnamed ratio	jump per named read over jump per unnamed read	recipients named in messages; reads	talk 0.079 vs 0.004 per read ($\times 19$) (H67); content 0.077 vs 0.016 ($\times 5$) (H50)	addressing is the coupling channel	ratio $\rightarrow 1^P$
Call-clock elasticity η	elasticity of the reply hazard to call duration (0 call clock, 1 wall clock)	call start and end times	$\eta_{\text{rep talk}} = -0.11$ [-0.16 , -0.06], 4 periods; regime I 0.11 [0.41, 0.61] (H40)	whether faster calls mean faster coupling	$\eta > 0.5^P$
Field share of co-activation f	share of excess co-activation gain from the scheduler’s start and stop	1-min activity spins, trimmed to the all-present window, block-shift null	0.72 median (8 periods), 0.83 after trim; regime I 0.11 (H38, fragile); shared pair covariance ϕ 0.04 [0.01, 0.06] (H86)	synchrony is the schedule, not coordination	$f < 0.5^P$
Kickoff alignment	rank of the day-1 content centroid against its own kickoff and 32 decoy kickoffs	statement embeddings, kickoff texts	top-1 in 18/33 periods (bge), 20/33 (gte) (H54); topic alarm fires in 30 min on 82–85% of kickoffs, 7–9% of placebos, AUC 0.92–0.94 (H36)	whether the goal landed	H36 alarm (measured)
Well depth $\Delta\rho$	extra loss of memory of agents’ positions at a kickoff, beyond an ordinary night	agent content vectors	$+0.26 \pm 0.04$ (s.e.), 16/18 kickoffs; along the kickoff 0.63, across 0.25 (H97); day-1 overshoot, no undershoot $U = -0.006$ [-0.020 , 0.008] (90%) (H125)	how hard a goal resets the swarm	$\Delta\rho \approx 0^P$
Kick and well times $1/\gamma_k$, $1/\gamma_s$	decay of the read response and of the agent’s own autocorrelation	content per call; reads	kick ≈ 7 calls, well ≈ 100 calls (≈ 1 h), ratio 14.8 [7.2, 30.4] (90%), #51 only (H130); post-erasure re-reading tail 8.2 [5.8, 11.8] calls (H44)	how long a message acts; how fast an agent forgets	ratio $< 2^P$
Top eigenvalue vs edge; PR	λ_1 over the 95th percentile of surrogate λ_1 ; participation ratio $(\sum \lambda)^2 / \sum \lambda^2$	agent $\times$ time matrices of activity, talk, content	above edge: talk 21/24 units, content 24/24, activity 7/24 (H12, fragile); PR -25% during #12 debate motions; spiked threshold $\ell > \sqrt{N/T}$ not measured	how many collective modes exist; a uniform mode is a field	activity mode above trimmed edge ^P
Room domain contrast	content separation of rooms beyond random regroupings after removing member constants and field directions	room membership; content	Q_{spont} 1.8–5.7, significant in 5/7 periods (bge) (H100); cross/within coherence ≤ 0.3 in 6/9 (H47)	rooms are domains; each goal rebuilds them	—
Context value per bit κ_C	$\Delta V_C/I_C$: commits lost per bit of allocation information erased	forced erasures; work commits per 20 calls	5.2 [3.8, 7.9], 9 periods pooled (G38 0.87, G51 7.0) (H87); -39% [-42 , -35] commits over 10 calls (H15); 20-call cap gives 0.88 [0.85, 0.91] of the output per call of a 40-call cap (H44)	the price of an erasure or a shorter cap	—
Day-end excess EP	share of debiased excess entropy production in the first and last 30 min of the day	behaviour-state fluxes per agent	0.80–1.00 of the excess in 12–25% of the time (G38, G40, G51) (H76)	irreversibility sits at the scheduler’s edges	excess outside edges ^P
Idea branching $\bar{R}$	mean new adopters per adopter of an idea marker	idea markers, reads	0.09–0.40, median 0.22; upper CI < 1 in 32/32 periods; spread across ideas $\text{CV}^2 \approx 1$ (H34)	how far an idea spreads	upper CI $\geq 1^P$
Change monitor	three-channel daily monitor (schema diff, counters, topic)	log schema, counts, embeddings	false alarms 0.085 [0.04, 0.17] per day; hits kickoffs 0.57, room changes 6/6 (H74)	dates platform and drive changes	measured
Dynamic mode lifetimes	lifetimes τ_k of a VAR(1) or OU fit of agent content	content per call or day	not measured; only one slow mode, $\tau \approx 23$ –28 d, regime I (H81)	what persists, beyond what co-moves	—

Appendix C: The full hypothesis × physics-model table

Figure 22 shows every scored hypothesis against the 17 physics models. Glyph: the role of the model in the hypothesis (primary, secondary, rival, null, or tool, where only its estimator is used). Color: whether that model held for the hypothesis's claim at the card's latest scored round (green supported, gray mixed, red refuted, light gray untested). Models 16 and 17 were added on 2026-10-07; the cards that test them (H97, H125, H127, H130 for 16; H12, H91, H92 for 17) carried model 11 or 01 before, which now shows as a secondary role. Figure 10 shows how one cell resolves into verdicts per goal period.

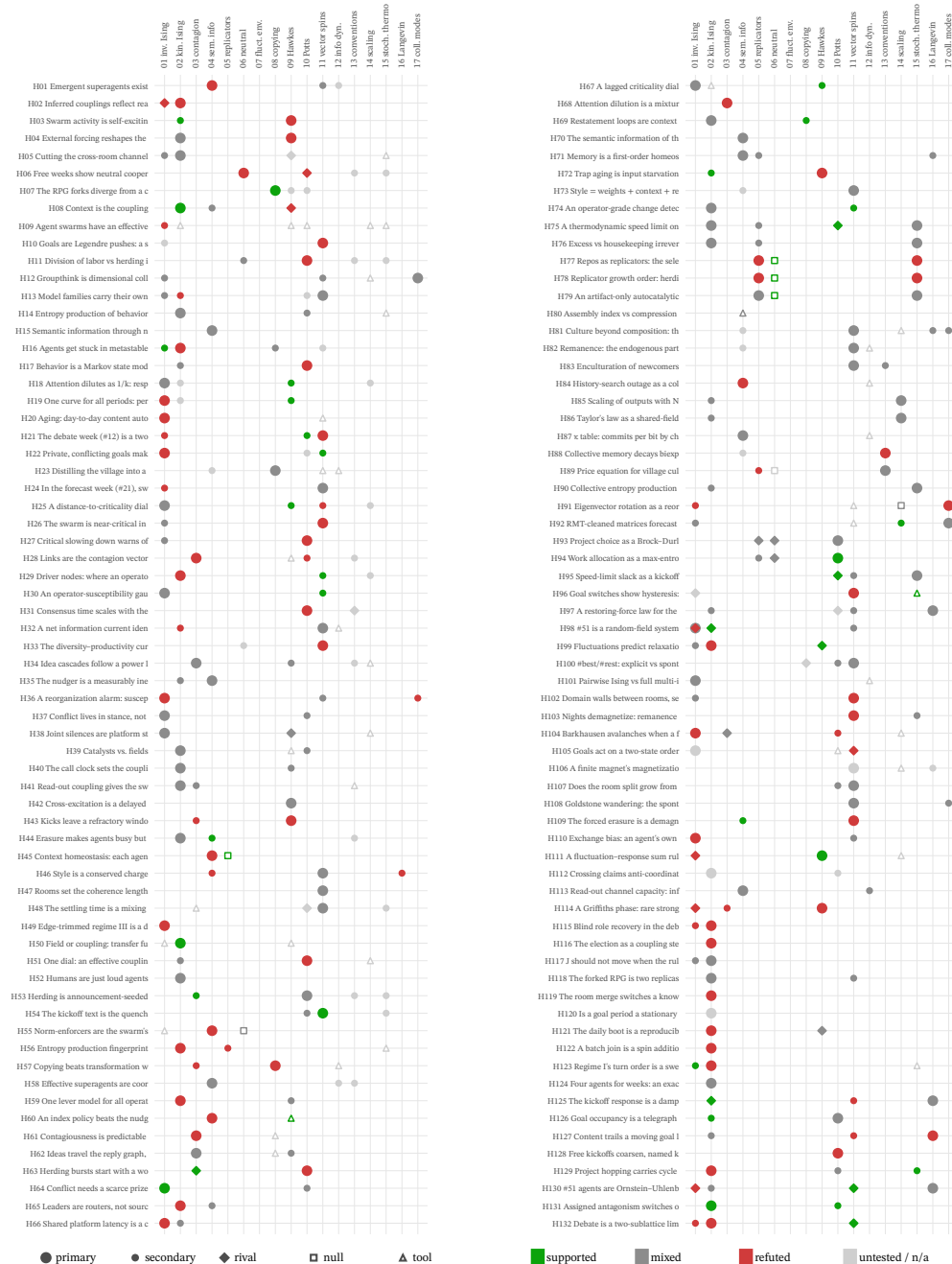


FIG. 22. All 132 scored hypotheses (rows) against the 17 physics models (columns). Look down a column to see how a model fared across hypotheses.

Appendix D: Every hypothesis on every goal period

Figures 23 and 24 show the full record: each of the 142 hypotheses (rows) on each goal period and natural experiment (columns) that it was tested on: 2875 period entries, of which 141 are n/a or pending. The rows are grouped by the primary physics model of each card, in the order of Secs. IV–VI. A cell is the verdict of that period’s test: its round-2 verdict where one exists, else round 1b, else round 1 (the same rule as `hypotheses/OVERVIEW.md`). “Mixed” includes tests that were inconclusive. Cells with a black outline are confirmatory tests on reserved data; shaded columns are reserved goal periods. Cards in gray type (H133–H142, run on 2026-10-07) are not yet scored. A row of mostly green cells is not by itself a supported claim: some cards pass in many periods and fail their strongest null (Sec. III C).

Appendix E: The most useful claims

Table VIII gives the 20 claims with the highest estimated usefulness among those with credence of at least 0.6, in short form. Appendix F lists all scored hypotheses by EU, with their faithfulness (p , M) and value (V). The scored claim for each is on its two-page summary in the compendium. Scores are the current values (round 2 where it exists).

TABLE VIII. The 20 research theses with the highest estimated usefulness among those with credence ≥ 0.6 (Claude-judged, Sec. III). p : credence (faithfulness); V : value if true (0–5); $EU = pV$; M: mechanism depth, how far the test reaches beyond a fit (M0: a pattern that repeats across goal periods; M1: the model’s own signature is predicted and a rival model fails; M2: in addition, a natural experiment agrees and synthetic data recover the effect); *fragile. No thesis has passed a confirmation test on reserved data.

ID	thesis (scored claim, short form)	p	V	EU	M
H44	Regime III (9 periods): a forced context wipe gives a one-call re-reading spike, ~ 8 -call tail ($\Theta_c + 0.106$ [0.089, 0.122]), 20–32% write dip; a 20-call cap costs 12% output per call.	0.85	3.50	2.98	M1
H38	Joint silences are operator-scheduled, not platform stalls; in regime III 70–80% of co-activation is the runner’s daily start and stop (f 0.72 median over 8 periods, 0.83 after trimming).	0.90*	3.25	2.93	M2
H46	Regime III: style is an identity charge plus context-held offset that forced erasures redraw (T 0.546 [0.522, 0.564], content flat); it identifies agents across goal switches at 0.85 (chance 0.15).	0.79	3.25	2.57	M1
H54	Over 33 kickoffs naming a shared target, day-1 content identifies its own kickoff (top-1 18/33 bge, 20/33 gte); #51 agents sit on private goals; human messages pull readers by $\Delta \approx 0.09$.	0.90	2.75	2.48	M1
H69	Regime III: restatement loops copy what the context still holds (in-context OR 3.38 [2.56, 4.46], 4/4 periods); an erasure ends them as a step, not a dose.	0.88	2.75	2.42	M1
H16	In #51, a forced erasure inside a pause-chain trap raises escape at next wake from 0.47 to 0.84 (log OR +2.68 [2.31, 3.16]); directed reads work by address, not dilution.	0.68	3.50	2.38	M2
H02	Activity-timing pairwise Ising couplings sit at the null floor (held-out wins in 3/21 chunks); the regime-III collective coupling was the operator’s day edges; the reserved #45 leader test failed.	0.94*	2.50	2.35	M1
H08	Naming and replying jump at the call that receives the message (mention 14/17, reply 17/17 periods); content moves toward it only within addressed replies (G51 +1.25 [0.98, 1.60] $\cos \times 100$).	0.85	2.75	2.34	M1
H67	In regime III the read-out talk loop gain is subcritical, g_{lag} 0.13 (25 units), matching Fano-implied gain 0.149 [0.123, 0.175]; regime I has no read-out gain on the chat clock.	0.87	2.62	2.28	M1
H42	In regime III, a read naming the recipient raises its talk at the read-out call by 0.076 [0.066, 0.086] (26 units); the named kernel survives a fitted Cox field (0.90).	0.80	2.75	2.20	M1
H74	A three-channel monitor with leave-one-period-out thresholds has per-day false-alarm rate 0.085 [0.04, 0.17], hits 0.57 of goal kickoffs and 6/6 room changes.	0.72*	3.00	2.16	M0
H52	Matched on salience, human messages get about twice the agent reply rate (+0.030 [0.019, 0.042], 17 periods); agent leaders share the premium (not deference).	0.60	3.50	2.10	M1
H29	Content influence depends on address (named 0.05–0.08 per message, unnamed ≈ 0 in #51); content-pull driver rankings fail held out; a reply-graph ranking predicts held-out spread weakly (0.23 bge, 0.12 gte).	0.65	3.17	2.06	M1
H40	In regimes II–III agents couple per call: a talk call’s reply hazard barely depends on wall time its last call spanned (η -0.11 [-0.16 , -0.06]); the wall clock is excluded.	0.82	2.50	2.05	M1
H36	A topic-shift alarm fires within the first 30 min of talk on 82–85% of 33 goal kickoffs against 7–9% of placebo day starts (AUC 0.92–0.94), robust across both embedding models.	0.74*	2.75	2.04	M0
H28	Links co-move with project switches in attention and work (HR ≈ 2.6) but mark bursts, not cause them: recipients switch at $6.5\times$ baseline before they can read the link.	0.81	2.50	2.03	M1
H50	In regime III, content couples at the recipient’s read-out call, as talk does: jump 0.033 [0.027, 0.039] (bge), CI > 0 in 10/16 units, none below; named messages $\times 5$.	0.79	2.50	1.98	M1
H43	Kicks leave no refractory window: a second mention read by a later call within 15 min keeps $R = 0.78$ [0.67, 0.89] of the first mention’s effect.	0.69	2.75	1.90	M1
H11	Herdling is not stigmergic: in 24 attention units, read chat mentions predict joins (OR 3.0) better than others’ commits (1.26; held-out ΔLL -0.047 [-0.070 , -0.024]).	0.75	2.50	1.88	M1
H72	In #51 trap aging is carried by the context’s self-share, not input starvation: own idle share carries 65% [59, 71] of the aging clocks’ held-out information (starvation -1%).	0.65	2.75	1.79	M1

Every hypothesis on every goal period it was tested on (part 1 of 2)



FIG. 23. Every hypothesis on every goal period, part 1: the useful models 02, 09, 11 and 01. Columns: goal periods by regime, then natural experiments (NE).

Every hypothesis on every goal period it was tested on (part 2 of 2)



FIG. 24. Every hypothesis on every goal period, part 2: the useful models 16 and 17, the partly useful models, the rest, and the framework cards.

Appendix F: All hypotheses by estimated usefulness

All 132 hypotheses, sorted by estimated usefulness $EU = pV$. p : credence (faithfulness); V : value if true (0–5); M: mechanism depth, how far the test reaches beyond a fit (M0: a pattern that repeats across goal periods; M1: the model’s own signature is predicted and a rival model fails; M2: in addition, a natural experiment agrees and synthetic data recover the effect); *fragile; verdict: the hypothesis as originally posed, at its latest round. The scored claim for each is on its two-page summary in the compendium.

TABLE IX. Hypotheses ranked by estimated usefulness, ranks 1–44.

#	ID	hypothesis	p	V	EU	M	verdict
1	H44	Erasure makes agents busy but unproductive	0.85	3.50	2.98	M1	mixed (no thrash)
2	H38	Joint silences are platform stalls	0.90*	3.25	2.93	M2	refuted (stalls) / supported (edges)
3	H46	Style is a conserved charge	0.79	3.25	2.57	M1	refuted (conservation)
4	H54	The kickoff text is the quench target	0.90	2.75	2.48	M1	supported
5	H69	Restatement loops are context fixed points	0.88	2.75	2.42	M1	mixed
6	H16	Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier	0.68	3.50	2.38	M2	mixed
7	H02	Inferred couplings reflect real influence	0.94*	2.50	2.35	M1	failed
8	H08	Context is the coupling	0.85	2.75	2.34	M1	supported
9	H67	A lagged criticality dial	0.87	2.62	2.28	M1	failed (lagged > equal-time)
10	H42	Cross-excitation is a delayed step at the next read-out	0.80	2.75	2.20	M1	mixed
11	H74	An operator-grade change detector	0.72*	3.00	2.16	M0	mixed
12	H52	Humans are just loud agents	0.60	3.50	2.10	M1	mixed
13	H29	Driver nodes: where an operator message moves the whole swarm	0.65	3.17	2.06	M1	not supported (drivers)
14	H40	The call clock sets the coupling	0.82	2.50	2.05	M1	failed (pooled P1) / supported in II–III
15	H36	A reorganization alarm: susceptibility and multi-information peak at transitions	0.74*	2.75	2.04	M0	mixed (bge) / failed (gte)
16	H28	Links are the contagion vector of herding	0.81	2.50	2.03	M1	not supported (contagion)
17	H50	Field or coupling: transfer functions and lag tell which	0.79	2.50	1.98	M1	mixed
18	H43	Kicks leave a refractory window	0.69	2.75	1.90	M1	refuted
19	H11	Division of labor vs herding is the sign of a Potts coupling, set by the goal	0.75	2.50	1.88	M1	mixed
20	H72	Trap aging is input starvation	0.65	2.75	1.79	M1	mixed
21	H58	Effective superagents are coordinated agents plus their artifacts	0.71	2.50	1.77	M1	failed (superagents)
22	H32	A net information current identifies de facto leaders	0.78	2.25	1.76	M0	failed (leaders)
23	H70	The semantic information of the artifact store	0.70	2.50	1.75	M1	failed (artifact semantic information)
24	H03	Swarm activity is self-exciting, and its criticality is set by the goal’s coupling mode	0.86	2.00	1.72	M0	failed (criticality)
25	H120	Is a goal period a stationary NESS? Within-period drift of J and EP (#38, #51 main)	0.68	2.50	1.71	M0	failed
26	H57	Copying beats transformation when the backlog is large	0.57	3.00	1.71	M0	refuted
27	H25	A distance-to-criticality dial, T/T _c , per window and per channel	0.85	2.00	1.70	M0	mixed
28	H34	Idea cascades follow a power law whose exponent reads off the distance to criticality	0.75	2.25	1.69	M1	mixed
29	H39	Catalysts vs. fields	0.67	2.50	1.68	M1	mixed
30	H47	Rooms set the coherence length	0.66	2.50	1.65	M1	mixed
31	H45	Context homeostasis: each agent keeps its context near a set point	0.82	2.00	1.64	M1	refuted
32	H53	Herding is announcement-seeded nucleation	0.59	2.75	1.62	M1	failed as posed
33	H73	Style = weights + context + register	0.80	2.00	1.60	M1	mixed
34	H85	Scaling of outputs with N	0.64	2.50	1.60	M0	mixed
35	H87	κ table: commits per bit by channel	0.63	2.50	1.57	M1	mixed
36	H56	Entropy production fingerprints the platform	0.69	2.25	1.55	M0	refuted
37	H12	Groupthink is dimensional collapse: few collective modes, shrinking participation ratio	0.77*	2.00	1.54	M1	failed (collapse)
38	H86	Taylor’s law as a shared-field gauge	0.68	2.25	1.53	M1	mixed
39	H13	Model families carry their own fields, and couple by family	0.74	2.00	1.48	M1	refuted (fields as positions)
40	H26	The swarm is near-critical in what it says but subcritical in when it acts	0.84*	1.75	1.47	M1	failed (R2) → unresolved
41	H76	Excess vs housekeeping irreversibility	0.73	2.00	1.46	M1	mixed
42	H15	Semantic information through natural scrambles: what keeps agents and the swarm viable	0.72	2.00	1.44	M1	mixed
43	H75	A thermodynamic speed limit on re-allocation	0.47	3.00	1.41	M1	failed as posed (field-limited slack)
44	H06	Free weeks show neutral cooperative dynamics with a cooperator core	0.80	1.75	1.40	M1	refuted

TABLE X. Hypotheses ranked by estimated usefulness (continued), ranks 45–88.

#	ID	hypothesis	p	V	EU	M	verdict
45	H62	Ideas travel the reply graph, not the room	0.70	2.00	1.40	M0	mixed
46	H01	Emergent superagents exist	0.62	2.25	1.40	M1	refuted (R4-R8)
47	H05	Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks	0.56	2.50	1.40	M1	inconclusive
48	H10	Goals are Legendre pushes: the response to a goal is predictable from unforced fluctuations	0.79	1.75	1.38	M1	failed (Legendre)
49	H31	Consensus time scales with the interaction graph's spectral gap	0.55	2.50	1.38	M0	failed
50	H68	Attention dilution is a mixture of two strategies	0.76	1.75	1.33	M1	refuted
51	H111	A fluctuation–response sum rule for talk: Fano factor = $1/(1 - g)^2$	0.65	2.00	1.30	M1	supported
52	H18	Attention dilutes as $1/k$: response to a message falls with the messages waiting at the turn	0.64	2.00	1.29	M1	mixed
53	H79	An artifact-only autocatalytic set	0.72	1.75	1.26	M1	mixed
54	H65	Leaders are routers, not sources	0.63	2.00	1.26	M1	mixed
55	H09	Agent swarms have an effective thermodynamics	0.56*	2.25	1.26	M1	mixed
56	H49	Edge-trimmed regime III is a dilute ferromagnet	0.81	1.50	1.22	M1	refuted
57	H80	Assembly index vs compression as an agent signature	0.68	1.75	1.19	M1	mixed
58	H59	One lever model for all operator inputs	0.59	2.00	1.18	M1	failed (one lever)
59	H71	Memory is a first-order homeostat	0.74	1.50	1.11	M1	mixed
60	H96	Goal switches show hysteresis: a coercive field	0.74	1.50	1.11	M0	mixed
61	H82	Remanence: the endogenous part of the mean field	0.73	1.50	1.09	M1	mixed
62	H35	The nudger is a measurably inefficient Maxwell demon	0.61*	1.75	1.07	M1	mixed
63	H81	Culture beyond composition: the emergent slow mode	0.61	1.75	1.07	M0	supported (regime I)
64	H92	RMT-cleaned matrices forecast tomorrow's alignment	0.71	1.50	1.06	M1	mixed
65	H100	#best/#rest: explicit vs spontaneous symmetry breaking	0.71	1.50	1.06	M0	mixed
66	H21	The debate week (#12) is a two-sublattice antiferromagnet	0.70	1.50	1.06	M1	supported
67	H97	A restoring-force law for the kickoff quench	0.70	1.50	1.06	M1	supported
68	H33	The diversity–productivity curve is an inverted U: the swarm's operating point	0.70	1.50	1.05	M0	failed
69	H89	Price equation for village culture	0.78	1.25	0.98	M1	mixed (no egogore attractor)
70	H63	Herd bursts start with a work signal	0.56	1.75	0.98	M1	failed (work signal)
71	H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?	0.39*	2.50	0.98	M1	mixed
72	H37	Conflict lives in stance, not topic: stance spins are antiferromagnetic	0.65	1.50	0.97	M0	mixed
73	H61	Contagiousness is predictable at first use	0.64	1.50	0.96	M0	failed (forecastability)
74	H64	Conflict needs a scarce prize	0.64	1.50	0.96	M1	supported
75	H113	Read-out channel capacity: information per call grows as $k^{1-\beta} \approx k^{0.34}$	0.64	1.50	0.96	M1	mixed
76	H130	#51 agents are OU particles in private wells: reads kick them; kicks decay at the well's rate	0.64	1.50	0.96	M1	failed
77	H95	Speed-limit slack as a kickoff-specificity gauge	0.35	2.75	0.96	M1	failed as posed
78	H99	Fluctuations predict relaxation (mean-field Glauber)	0.63	1.50	0.95	M1	supported
79	H88	Collective memory decays biexponentially	0.62	1.50	0.93	M0	mixed
80	H109	The forced erasure is a demagnetizing pulse: where is the room's order stored?	0.62	1.50	0.93	M1	failed
81	H94	Work allocation as a max-entropy equilibrium	0.61	1.50	0.92	M1	supported
82	H51	One dial: an effective coupling collapses the phase diagram	0.61	1.50	0.91	M1	failed
83	H04	External forcing reshapes the response kernel, reversibly	0.49*	1.83	0.90	M1	mixed
84	H124	Four agents for weeks: an exact kinetic Ising benchmark for mean-field approximations (#4, #6)	0.59	1.50	0.89	M1	mixed
85	H127	Content trails a moving goal like an overdamped particle: agents that call more catch up faster	0.59	1.50	0.88	M1	failed
86	H66	Shared platform latency is a common-noise field	0.58	1.50	0.87	M1	refuted as posed
87	H24	Forecast week (#21): switching from independent drafts to comparison switches the coupling on	0.42*	2.00	0.84	M1	failed (switch-on)
88	H119	The room merge switches a known adjacency on and off: recover it from the dynamics (NE42)	0.54	1.50	0.80	M1	failed

TABLE XI. Hypotheses ranked by estimated usefulness (continued), ranks 89–132.

#	ID	hypothesis	p	V	EU	M	verdict
89	H07	The RPG forks diverge from a common ancestor at a measurable rate	0.52	1.50	0.78	M1	mixed
90	H117	J should not move when the rules do: the mid-goal reset in #32 (NE35)	0.52	1.50	0.78	M0	mixed
91	H48	The settling time is a mixing time on the read-out graph	0.51	1.50	0.77	M0	inconclusive
92	H102	Domain walls between rooms, set by bridging agents	0.51	1.50	0.77	M0	failed
93	H27	Critical slowing down warns of herding waves hours ahead	0.43	1.75	0.75	M0	failed
94	H104	Barkhausen avalanches when a field is stepped	0.49	1.50	0.74	M0	failed
95	H20	Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag	0.72	1.00	0.72	M0	mixed, leaning refuted
96	H41	Read-out coupling gives the swarm a light cone	0.36*	2.00	0.72	M0	mixed
97	H19	One curve for all periods: loop gains collapse onto one function of a control parameter	0.71	1.00	0.71	M0	failed
98	H91	Eigenvector rotation as a reorganization signal	0.69	1.00	0.69	M0	failed
99	H77	Repos as replicators: the selection-resolution bound	0.55	1.25	0.69	M0	failed (selection resolution)
100	H78	Replicator growth order: herding vs division of labor	0.55	1.25	0.69	M0	failed (growth order)
101	H123	Regime I's turn order is a sweep: equilibrium-looking statistics, nonzero entropy production	0.67	1.00	0.67	M1	failed
102	H60	An index policy beats the nudger	0.44	1.50	0.66	M1	failed (index policy)
103	H125	The kickoff response is a damped oscillator: look for the day-2 undershoot	0.64	1.00	0.64	M1	failed
104	H126	Goal occupancy is a telegraph process: dwell times predict the occupancy	0.64	1.00	0.64	M1	failed
105	H14	Entropy production of behavior sequences: per-family arrows of time, collective irreversibility	0.82*	0.75	0.61	M0	mixed
106	H17	Behavior is a Markov state model with few metastable sets	0.61	1.00	0.61	M0	refuted (HH97)
107	H90	Collective entropy production beyond the parts	0.41	1.50	0.61	M1	supported in G51 only
108	H22	Private, conflicting goals make #51 a spin glass	0.60	1.00	0.60	M0	failed
109	H103	Nights demagnetize: remanence decays per night	0.59	1.00	0.59	M0	failed
110	H55	Norm-enforcers are the swarm's immune cells	0.57*	1.00	0.57	M0	failed
111	H93	Project choice as a Brock–Durlauf equilibrium	0.56	1.00	0.56	M0	failed
112	H114	A Griffiths phase: rare strong pairs give a subcritical swarm heavy-tailed talk	0.54	1.00	0.54	M1	failed
113	H83	Enculturation of newcomers	0.53	1.00	0.53	M0	mixed
114	H118	The forked RPG is two replicas of one dynamics: damage spreading between rooms (NE15, #35)	0.52	1.00	0.52	M0	mixed
115	H121	The daily boot is a reproducible quench: hundreds of replicas of relaxation to the steady state	0.52	1.00	0.52	M0	failed
116	H129	Project hopping carries cycle currents: the sticky Potts walker breaks detailed balance	0.52	1.00	0.52	M0	failed
117	H128	Free kickoffs coarsen, named kickoffs freeze: a kinetic Potts quench	0.51	1.00	0.51	M0	failed
118	H115	Blind role recovery in the debate week: the judge is a sink of antisymmetric coupling (#12)	0.50	1.00	0.50	M0	failed
119	H98	#51 is a random-field system	0.49	1.00	0.49	M0	failed
120	H122	A batch join is a spin addition: do incumbents respond as fitted couplings say? (NE27, NE33)	0.47	1.00	0.47	M0	failed
121	H131	Assigned antagonism switches off one read-out after the prize goes: a field quench	0.47	1.00	0.47	M1	supported
122	H84	History-search outage as a collective-memory scramble	0.42	1.00	0.42	M0	failed
123	H101	Pairwise Ising vs full multi-information in co-usage	0.42	1.00	0.42	M0	mixed
124	H116	The election as a coupling step: influence should flow toward the winner after the result (#26)	0.42	1.00	0.42	M0	failed
125	H23	Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans	0.37	1.00	0.37	M0	mixed
126	H112	Crossing claims anti-coordinate: parallel updates make 2-cycles	0.44	0.50	0.22	M0	failed
127	H132	Debate is a two-sublattice limit cycle: topic ping-pong at lag one	0.44	0.50	0.22	M0	failed
128	H108	Goldstone wandering: a spontaneous room direction should drift, a fielded one stay pinned	0.42*	0.50	0.21	M0	supported
129	H106	A finite magnet wanders as 1/N: the slow mode's decay time should grow with village size	0.62	0.30	0.18	M0	mixed
130	H105	Goals act on a two-state order parameter	0.36	0.50	0.18	M0	mixed
131	H107	Does the room split grow from zero (an instability) or appear at once (a hidden field)?	0.35	0.50	0.17	M0	mixed
132	H110	Exchange bias: an agent's own artifact shifts its goal-switch loop	0.35	0.50	0.17	M0	mixed